A Local Grammar for Proper Names

Hayssam N Traboulsi

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University of Surrey

Information Extraction and Multimedia Group
Department of Computing
School of Electronics and Physical Sciences
University of Surrey
Guildford, Surrey GU2 7XH, UK

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Abstract

Conventionally, the identification of proper names in arbitrary texts is accomplished by using a surface linguistic approach or a probabilistic approach. The need to use costly tagged corpora is inevitable to both approaches. In addition, the visual examination of proper name surrounding contexts in such corpora is vital in order to utter their local grammar. In this research, we investigate how techniques in corpus linguistics, including frequency, collocation and concordance analyses, can be used to automatically identify the local grammar of proper names in untagged corpora.

If biographical dictionaries are considered to be a source of knowledge about notable men and women recorded in history, daily news stories might also be regarded as a source of knowledge about personalities currently shaping our world. The local grammar of proper names discussed in this work can be used to automatically acquire such knowledge from journalistic texts.

Analysis conducted on news data leads to the observation that there are undoubtedly repetitive patterns used in journalism when describing important personas. These patterns suggest the possibility of a hands-on wealth of both lexical and world knowledge of proper names and making it available for computational consumption.

Our approach to handling proper names in news texts is implemented in a system called NExtract. This system comprises three major units: a lexical analyser, a sublanguage parser and a proper name classifier. The key component of NExtract is its sublanguage parser which uses cascades of finite state transducers (implementing the discovered local grammar) to achieve syntax-semantic mapping. We present a small-scale evaluation of NExtract's performance on samples of unseen news texts, which is held to support the validity of our approach to handling proper names.

Email: csp1ht@eim.surrey.ac.uk

WWW: http://www.eim.surrey.ac.uk/

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Chapter 1

1 Preamble

1.1 Introduction

The automatic analysis of natural language texts has in recent years covered a range of theoretical and practical enterprises. One key hybrid enterprise is that of Information Extraction (IE); a Natural Language Processing (NLP) technology that locates specific pieces of information or *events* in free text and uses it to fill in different kinds of template. In IE, a body of texts (usually newswire articles, newspaper or journal articles etc.) is the input, while the output is a set of slots that may contain attributes of one entity, a relationship between entities, or an event with various entities playing roles and/or being in certain relationships. Entity attributes may include proper names of people, organisations or locations. The IE subtask of identifying such names in arbitrary texts is defined as Named Entity (NE) recognition.

Despite the availability of large lexical systems, the analysis of naturally occurring texts is still not an easy task due to the ever growing number of new words arising from new coinages, slang expressions, and especially *proper names*. The latter are of particular concern; proper names fall partly inside and partly outside the lexicon and grammar of a language and are frequent in an arbitrary text; in addition to IE, the analysis of proper names is important to other text processing applications: *Machine Translation*, *Information Retrieval* and *Question Answering Machines*.

Two main approaches are generally adopted for the identification of proper names: A surface linguistic approach or a probabilistic approach. Either approach has achieved a combined score of *precision* and *recall* comparable to human score ($\approx 90\%$ on news texts), but at the high cost of manual effort to tag corpora needed to compile grammar rules for the linguistic approach, and train language models for the probabilistic approach. In this research, we attempt to reduce the high cost of rule compilation necessary for NE

extraction systems using a linguistic approach. We present a corpus-based method to identify patterns of proper names. This method involves understanding the behaviour of proper names in free texts: by first identifying lexical tokens that occur in the neighbourhood of proper names; second using the behaviour to find out if there are patterns of usage; and third, using the frequent patterns in a simple NE extraction system which we have named (NExtract).

NExtract is developed in Java to extract person and organisation names from news articles. It is a rule-based system that exploits behaviour (frequent patterns) of proper names learnt from a large specific domain corpus in favour of acquiring lexical and world knowledge of the names it encounters.

1.2 The Challenge of Proper Names

Proper names are defined as names of persons, places, and dates, which "may also be expected to lack number contrast, determination and modification" [1]. However, there are exceptions, usually found in general language texts, like *the Simpsons* (proper name with number mark), *The Thames* (determination) and boring *Guildford* (modification) [1]. Proper names can be viewed as labels, which have no apparent meaning, used to refer to different entities and distinguished by the initial capitalised letter or the fact that they are not translated. However, nouns such as *sun*, *earth* or *moon*, semantically covered by the rules of proper names, are always translated and regarded as common nouns. In contrast, proper names, especially names of famous people, are used, particularly in metaphorical contexts, as common nouns, as in *I read some Shakespeare* or *poems of Shakespeare*. In addition, adjectives derived from proper names like *Elizabethan*, *Belgian*, *German*, *Brazilian*, or *British* are not considered as proper names despite the fact they are initially capitalised [2]. One more point to add here is that professional titles, in free texts, are occasionally capitalised although they are not considered as proper names.

Proper names can occur in variant forms; names such as *George W. Bush*, *Bush* and *President Bush* are all used to refer to the current and former presidents of the United States of America. We can talk about the *United States of America* and *the States*, *Austin Reed* and *Austin Reed Group plc*. The *variants* phenomenon of proper names adds

another challenge to IE systems aimed at the identification and extraction of names and the entities they refer to [3]. In addition, it confirms the impossibility of the use of some kind of super dictionary as a solution to the proper name research challenge.

There is also the problem of semantic or entity-type ambiguity; the name *Bush* could refer to *George W. Bush* (the son), *George Bush* (the father) or *Bush*, a district in Los Angeles. Entity-type ambiguity is most likely to occur with proper names as persons are named after places (*George Washington* after *Washington*), places after famous people (*Vancouver* after the Pacific explorer *George Vancouver*), companies after their founders (*Austin Reed*). Moreover, this could be worse if we take into consideration those names which are unconventionally created; a car credit company named *Yes*[1] and a phone cards company called *Go Talk*[2] [4]. The arbitrary nature of proper names – any word can be a proper name – worsens the problem of lexical ambiguity already encountered in NLP. Words like *Bush*, *Major* and *Dolly* can occur with an initial capital without referring to proper names. Therefore, capitalisation rules are not sufficient enough to distinguish between what is a proper name and what is not.

Proper names show structural ambiguity in prepositional, conjunction scope and with respect to the possessive pronoun. A prepositional phrase may form part of a single name, like in *International Centre for Distance Learning*, or it may be independent of the preceding noun phrase like in *Tony Blair of England* where *of* splits two separate names - *Tony Blair* and *England*. Similarly, the conjunction *and* may conjoin two distinct names like *USA* and *England* or it may be part of a single name like the case of *Victoria and Albert Museum*. Finally, proper names can also occur in an apposition construction such as *England's Tony Blair* where the possessive pronoun relates two distinct names or *Holmes Place's* where the apostrophe (') and the letter (s) are part of the name [4].

Systematic metonymy[3] is an important issue one should be aware of when referring proper names to each other. For example, *Washington* refers to a place or to the *United*

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[1] http://www.yescarcredit.net.
[2] http://www.mobilemall.com.au
[3] A figure of speech in which one word or phrase is substituted for another with which it is closely associated, as in the use of *Washington* for *the United States government* or of *the sword* for *military power*. *(www.dictionary.com)*

States government; *Wall Street Journal* refers to the newspaper, and the company that creates it [4].

1.3 Existing Approaches and Systems

There are over thirty NE extraction systems for a variety of languages including English, Spanish, Chinese, Japanese and Thai. Machine learning (probabilistic), rule-based (linguistic) and hybrid approaches have been adopted in their development.

The probabilistic approach uses a language model trained on large manually pre-tagged corpora to learn patterns of identification. This approach results in good performance if trained with a rather large corpus. For example, Nymble achieved a high success rate (90%) in MUC-7 after manually tagging around half a million words [5].

The linguistic approach is based on rules describing the syntactic and lexical structure of language expressions containing proper names, e.g. *title + sequence of capitalised words* or *sequence of capitalised words + marker*. Hence, the exploitation of these rules most often requires resources like lists of markers (titles and trigger words such as *Mr.* and *Inc.*), gazetteers (large dictionaries of known proper names, e.g. place and first names) and general language dictionaries. These rules are either intuitively developed or manually compiled from previously tagged text samples. NetOwl of IsoQuest and FACILE of UMIST are examples of systems based on this approach [6], [7].

The hybrid approach is a combination of the probabilistic and linguistic approach. This approach is currently implemented in NE extraction systems developed by the University of Edinburgh (LTG), the University of Manitoba, and Cucchiarelli and Velardi [9], [8].

NE extraction systems evaluated in various MUCs have achieved good performance for specific texts (journalistic texts) where strict writing constraints are imposed. These systems, however, failed to maintain similar results when applied to informal texts such as e-mails and news messages. This failure is due to differences in the syntax of proper names occurring in informal and journalistic texts. For example, *trigger* and *marker* words like *Mr.*, *Mrs* and *Inc.* or *Ltd.* used with person and company names in journalistic

texts are almost missed in informal texts. These observations show that the grammar related to proper names is domain dependent [10]. Therefore, this grammar has to be revised each time proper names are to be identified in different kinds of documents. The manual compilation of a grammar related to proper names is known to be a lengthy and boring process [11].

This thesis presents a corpus-based method to automatically identify patterns of proper names used in texts. Once validated, the patterns can be used as rules for automatic recognition of proper names.

1.4 The Approach

In general, *keywords*, which are informative words (largely common nouns), embedded in the title or text of a document, are regarded as the terminology of one domain. However, there are instances where proper names are used either as eponyms (Fermi, Newton, etc.) or as names of cited authors in scientific texts where these names appear typically in proceedings or as references. Also, there are instances where proper names are used to describe the action of specific persons or institutions. A good example of this genre of text is a news text. The frequently used proper names in a news text belong to the special language of finance and commerce interspersed with names of people, key decision makers, currencies and financial institutions. For us these proper names can be chosen to indicate the main content of a given news article. Other instances of the use of proper names include political texts. However, the language of politics is not specified as the language of finance.

Financial news reports comprise statements of key person names and, typically, cues for finding proper names can be found surrounding saying verbs like *said*, *told* and *added*. Other important clues are frequently related to the often-contiguous appearance of the organisation name, which the reported person works for. Less frequent but contiguous items interspersed between the person and organisation names are the person's professional title and a modifier that defines the organisation range (e.g. *bookmaker*, *broker*, *private client stockbroker*, etc.). Table 1.1 below shows some examples of these statements (hence after patterns).

Table 1.1: Excerpts taken from Reuters' news stories.

Pattern	Excerpt
said PN, TITLE at ORG.	said James Laing, fund manager at Aberdeen Asset Management.
said ORG TITLE PN.	said Dresdner Bank spokesman Eberhard Seitz.
said PN, TITLE at MOD ORG.	said Freddie Tulloch, trader at bookmakers Financial Spreads.
said Title PN.	said campaign spokesman Jim Reynolds.

Despite the arbitrary nature of proper names and the apparent lack of linguistic knowledge about their lexico-semantic category, one can discern repetitive patterns associated with proper names: Names of persons, places and things cannot generally be preceded by determiners; in narrative texts names of people are usually preceded by the token *said* and followed by *told*. The high frequency of these patterns suggested to us the existence of a *local grammar* governing the use of proper names. Local grammar is a way of describing the syntactic behaviour of a group of individual elements which are related, but whose similarities cannot easily be expressed. Gross first used the notion of local grammar to capture expression of time and date as well as addresses in letters [12].

This thesis describes a corpus-based method for discovering such patterns and then inferring rules that help increase the accuracy of a named entity program. We examine the manner in which: (a) a small number of *keywords* in a domain are distributed throughout the corpus; and, (b) a *local grammar* that is idiosyncratic of a class of named entities (person and organisation names) in the domain governs the usage of the *keywords*. The term *local grammar* is used to suggest that the local grammar supplants the general/universal grammar when one is required to articulate information related to idiosyncratic entities.

The rather unique distribution of *keywords* in a specific domain acts as a signature of the authors of texts of the domain. A small number of high-frequency single keywords dominate many of the texts in a domain and many of these single keywords (i) are used in the formation of compound terms and (ii) have high collocation strength into other high-frequency keywords. We illustrate the development and efficacy of our method on financial news articles, which constitutes a specific domain.

The basic idea is that *local grammars* for proper names are compiled, which can then be used in the identification and classification of proper names. For instance, the local grammar that describes the first pattern in Table 1.1 is shown in Figure 1.1 below. This local grammar, represented in Finite State Automata (FSA), recognises person and organisation names and specifies the professional title of the person in that organisation. However, proper names that appear in isolation (out of *local grammars*) are disambiguated using lists of person first names and organisation markers (Ltd, Inc, AG, Plc and so forth).

Figure 1.1: FSA that recognises related person and organisation names.

With this technique we are hoping to analyse proper names occurring in Arabic news articles where named entities are not initially capitalised. We are also intending to take the notion of local grammar further and apply it to domains other than news articles. Our list of priorities includes scientific papers where, for example, abbreviations referring to organisations and chemical elements need to be disambiguated.

This work is implemented in a named entity extractor called NExtract. This system is implemented in Java to analyse news stories from which to extract person and organisation names.

1.5 Outline of the Thesis

This thesis is divided into five chapters. In Chapter 1 we give a general overview of the task of named entity extraction, define its importance for IE and expose typical challenges a computational analysis of these entities has to face. We then show different approaches and systems that have been utilised to extract proper names from natural language texts. Finally, we show the approach we have adopted to extract these entities from similar texts; news articles.

Chapter 2 reviews proper names from a number of perspectives. It roots proper names in the local culture and often shows the quirks of the language of the culture. The *onomastic*

approach of person and place names is also discussed, followed by a linguistic study of proper names including syntactic, semantic and pragmatic aspects. Apart from providing relevant literature within the field of proper names analysis, a review of various proper name extraction systems is also provided. These systems achieved high values of *precision* and *recall* only after manual tagging of a very large number of words. In addition, they are based on approaches that have been evaluated only for specific domains and languages; there is no concrete proof that similar results and performance may be achieved in other domains and languages. Therefore, more effort should be dedicated to address challenges related to the adaptability of linguistic resources to new languages and domains. This thesis presents a corpus-based method to generate rules for a rule-based named entity extractor with minimal manual effort.

In Chapter 3 we analyse the language used by journalists when reporting statements made by influential personalities in English news articles. This analysis suggests the existence of distinctive and consistent patterns which form the theoretical framework of our work.

Chapter 4 introduces the system architecture, explaining the specific functionality of each component in greater detail. Four case studies and relevant experiments performed on financial time series data are presented and the results evaluated. This chapter ends with a brief discussion about outstanding issues.

Conclusions, contributions and future work are presented in Chapter 5.

Chapter 2

2 A Note on Proper Names and their Behaviour

This chapter reviews the subject of proper names from different perspectives. We believe that a review of the various aspects of English proper names would bring insights to their analysis for text understanding. Here we study names in relation to nature, naming practices, and linguistic features together with state of the art computational work on proper name recognition.

2.1 The Nature of Proper Names

Proper names are language expressions that denote unique entities, typically including people, organisations, locations, commodities and so forth. They are rooted in the local culture and sometimes show idiosyncrasies of the language of the culture. This fact explains the difference in naming systems from one culture to another. The discipline that studies the origins and forms of proper names is known as *onomastics*. Historically, most work in *onomastics* has been in the areas of place and personal names.

The understanding of names and the investigation of their cultural insights, settlement history and linguistic features have been the principal aims of many organisations. The *American Name Society* and the *Canadian Society for the Study of Names*, for instance, are mainly dedicated to the study of place and personal names, but also proper names of all kinds [13] and [14].

Personal Names: *Anthroponymy* is the subdivision of *onomastics* that is only concerned with the study of the nature of personal names [15]. The word is derived from the Greek words *anthropos*, meaning “man”, “mankind” or “person” and *anoma*, meaning “name”.

In general, naming systems differ in the way they use the following types of *anthroponyms*:

Personal names: They are the first two underlined names in each of the following name examples: *Mary Alice Smith*, *Johannes Jakobus Venter*, *Nomusa Philomena Dlamini*, *Jacques Pierre de Robillard*. In most European languages, these are often called "Christian names" or "first names". However, both nominations can be confusing if we are talking of the personal names of non-Christian societies who have different first names like Muslim, Jewish and Buddhist, or who put the surname first like the Hungarian, Chinese and Japanese.

Surnames: These names are shared by members of a nuclear family, and in patrilineal societies where the surname passes on from son to son. Euro-Western surnames are greatly based on occupation (Cook, Gardiner, Wainwright, De Jaeger, Schumacher); on the name of the father (Johnson, Jackson, Mattieson, O'Neill, MacDonald); on location (Van der Berg, Van der Westhuizen, Van der Walt, Du Bois); or on physical characteristics (Swart, Van Rooyen, De Lange, Armstrong).

Clan names: A *clan* is a traditional social unit consisting of a number of families claiming a common ancestor, following the same hereditary chieftain, sharing identifying marks or slogans, associated with a particular geographical area and usually do not intermarry. The Scots among the British peoples and the Zulus among the South African peoples are divided into clans.

Patronyms: Names used to address or identify someone by referring to him or her as *son of so-and-so* or *daughter of so-and-so*. Many Euro-Western African surnames are originally father names and this naming practice is also common in Russia and Iceland. For example in Russian, *Ivan's* son would be called *Ivanovich*.

Teknonyms: In most parts of Africa and the Arab World, an extremely common way of identifying and addressing a person is to call him or her father of *so-and-so* or *mother of so-and-so*. Such names are called *teknonyms*. For example, *Abu Ali* suggests that the named person is the father of *Ali* in most of Arab countries.

Nicknames: This is a more informal way to name persons. These names are not normally added to formal documents of the named person. This type of name is frequently applied to children in many cultures. However, Palestinian leaders among Arabs are always given *nicknames* e.g. *Abu Ammar* is the *nickname* (nom-de-guerre) given to *Yasser Arafat* (the late Palestinian president).

Ethnonyms: Names used to refer to ethnic groups: Jew, Boer, Englishman, Arab, Indian, Russian, Slav, Zulu, Sotho, etc. These are often offensive. For example, *Wop* has been used as a disparaging term for a person of Italian birth or descent.

Place names: *Toponomy* is the subdivision of *onomastics* that mostly tries to trace the nature of place names. The word is originally derived from the Greek words *topos* meaning "place" and *onoma* meaning "name" [15]. The hierarchy of place names includes geographical entities like seas, mountains, rivers, and lakes as well as institutional entities such as streets, roads, houses and bridges. Similar to personal names, place names often reflect the history, beliefs, and values of a culture. American place names for example, are taken from the cultures of America's early explorers, i.e. Spanish, French, Dutch, Indian and English. Similarly, Celtic, Roman, Anglo-Saxon, Scandinavian and Norman names provide great insights into British history.

The naming process of places is very creative. Places can be named after persons who have a religious or historical meaning. For instance, all place names starting with *St*. or *Santa* are named after saints, e.g. *St Neots* in the UK and *Santa Cruz* in California. Other place names are named after Royal family members (e.g. *Victoria*) or after their discoverer or conqueror (e.g. *Alexandria* in Egypt and *Vancouver* in Canada). There are also place names indicating well-known events; a town near Hastings is called *Battle* after the famous battle of Hastings.

The study of the nature of proper names is a very broad and complex subject because of its relationship to many other subjects including history, archaeology, geography and, more importantly, linguistics. The following sections are dedicated to understanding linguistic, semantic and pragmatic aspects of proper names.

2.2 Linguistic Aspects of Proper Names

Computational linguistics applications always benefits from studies in theoretical linguistics. Therefore, a linguistic study of proper names would be of great importance for a computational system seeking the analysis of this type of noun. Early linguistic studies of proper names were only concerned with their meaning; the way they refer to entities and how they differ from common nouns. On the other hand, modern linguists tend to concentrate on the semantic and syntactic aspects of proper names. We suggest that we can learn much more from a modern linguistic analysis of proper names to serve our aim of building a computational model. This section will first investigate the syntactic structure of proper names and then advance to their semantic aspects.

2.2.1 Syntactic Features of Proper Names

Linguists define two essential syntactic features of proper names. The first involves their internal syntactic structures and the second concerns their external grammatical structures.

The Internal Structure of Proper Names: Quirk et al. (1985) defines two syntactic categories of proper names - names with and without articles. The latter includes personal names (e.g. *George W. Bush*), temporal names (e.g. *Christmas* and *Easter*) and place names (e.g. *London*). The former category, names with articles, is divided into three subcategories. This division is entirely based on the internal structure of these names. Thus, we may see names without modifications like *The Thames*, names with pre-modifications similar to the *English Channel* and finally names with post-modifications like *the House of Commons*. Figure 2.1 below illustrates Quirk's taxonomy of syntactic categories of proper names [1].

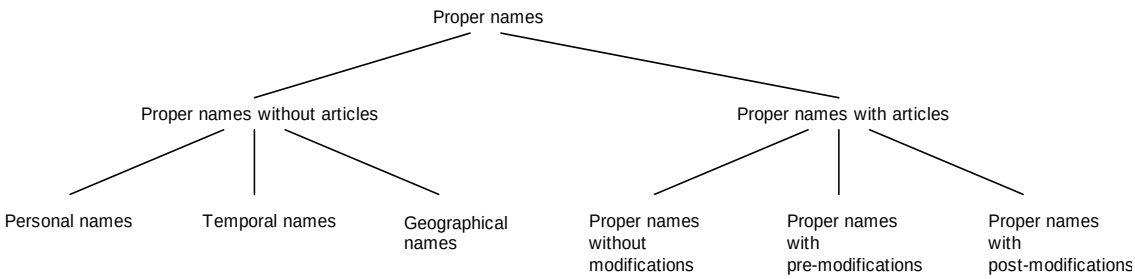

Figure 2.1: Quirk's taxonomy of syntactic categories of proper noun.

Allerton, concerned with the delimitation issues of proper name classes and the subcategories they might have [16], suggested four syntactic categories of proper names. These syntactic categories were further expanded into other subclasses as shown in Table 2.1 below.

Table 2.1: Syntactic categories of proper names (Allerton 1987).

Syntactic Category	**Syntactic Subcategory**[4]	**Example**
Pure Proper Names	Simple PN Article + Simple PN Title + Simple PN	*London* *The Hague* *Mr. George Bush*
Mixed Proper Names	PNs + CNs Article + PN + CN RA + PN PN + Article + RA or CN The + (adjective) + CN + of + PN	*Hyde Park, New York City* *The Suez canal* *Northern Island* *Henry the Eighth* *The University of Surrey*
Common-Based PN	The + adjective + head noun The + modifier noun + head noun Adjective + noun Modifier noun + head noun	*The Red Sea* *The Labour Part* *Central Park* *Salt Lake City*
Coded Proper Names	With the definite article *the* Without any article	*The FBI* *BBC, M25, Nato*

Proper names can occur in variant forms, commonly observed in real texts. Carroll thoroughly discussed this special feature of proper names [17], focusing on the relationship between an established place name and its variant forms. Interestingly, he observed that most place names have a *category word* and *namestem*. By category word he means the associated word that classifies the name into a particular category. Thus, the word *circus* in *Piccadilly Circus* determines its category (place name). The *namestem* is the part of the place name that distinguishes it from its counterparts in the same category, e.g. *Piccadilly* in *Piccadilly Circus*. He also observed that namestems are sometimes used to refer to the same place, e.g. *New York* instead of *New York City*. In this case they are called *nameheads*. However, not all place names can have nameheads. Carroll says that a namestem becomes a namehead only if it points to a unique and unambiguous referent. For example, the namestem *Piccadilly* in the above example cannot become a namehead because it does not stand for the whole name on its own.

Proper names also show syntactic polymorphy - the same name can have many forms. This is mainly due to transliteration across languages. For example, *Beirut* and *London* in

[4] PN: Proper Noun
CN: Common Noun
RA: Restrictive Adjective

French are *Beyrouth* and *Londres* respectively. However, proper names seem to be more stable than other parts of speech across languages. Allerton proposed that the more language specific a name is, the more prone it is to translation. This justifies the reason why temporal and organisation names are always translated and not, for example, personal names which are often transliterated [16].

The External Structure of Proper Names: Proper names normally function as single and rigid units with respect to grammar. This characteristic causes this class of nouns to behave very differently from common nouns. Quirk et al. (1985) make the observation that proper names normally lack article and number contrast. Hence, a place name like *London* cannot be *a London*, *the London* or have any plural forms, i.e. *Londons*. The insertion of modifiers is not allowed under any circumstances. Proper names can only be modified throughout relative clause and apposition non-restrictive modifiers. Consider for example:

> Dr Brown, *who lives next door*, came from Canada yesterday.
>
> Tony Blair, *the British Prime Minister*, arrived in Damascus Airport yesterday morning.

However, (Quirk et al., 1985:288-289) observed some behavioural similarity between proper and common nouns. Thus, they reported the following:

> *Plural forms* could be applied to proper names under metaphoric cases, e.g. *Shakespeares* to mean 'authors like Shakespeare'.
>
> *Surnames* take plural forms when occurring after the article 'the' in order to refer to the family or all the members of the family, e.g. the Clintons.
>
> *Articles, determiners* and *possessive pronouns*, when added to a proper noun, vary its meaning. For example, the possessive pronoun 'his' in a phrase like *Shakespeare* in *his new Shakespeare* varies the meaning of the word *Shakespeare* appearing after it into a copy of his book. When a proper name occurs after the indefinite article 'a', it denotes an entity called by this name.
>
> *Proper names* can be used as common nouns as well. Consider the following examples:

I know another *John who went to Oxford*.

I spoke to *the younger Mr Hamilton, not Mr Hamilton the manager*.

Finally, the aspect of the referent that a proper noun aims towards could be altered if the determiner 'the' is combined with restrictive modifiers. For example, the referents of the proper names in 'in England' and 'in the England of Queen Elizabeth' are not the same since the former denotes a place and the latter points to an era.

2.2.2 Semantics of Proper Names

The investigation of the semantics of proper names has always been of interest for linguists and psychologists as well as ancient philosophers.

Quirk et al. semantically define proper names as names of people (*Shakespeare*), places (*London*), months (*December*), days (*Sunday*), festivals (*Easter*), magazines (*Vogue*) and so forth. They remarked that proper names are much given to countable nouns (man, city or date) rather than uncountable nouns (sugar, bread or water).

Allerton has also considered the semantics of proper names. He argued that proper names have individualistic meanings [16]. That is, their meanings are separate and not integrated with other lexical meanings and have their own categories, which are defined according to the referents proper names point to. Allerton observed that being aware of names increases one's language ability and enhances general knowledge. Consecutively, this proposes that proper names are very informative and helpful for text understanding. Table 2.2 below illustrates the semantic categories of proper names suggested by Allerton.

Table 2.2: The semantic categories of proper names (Allerton , 1987).

Category	**Examples**
Human Beings	Aristotle, Socrates
Vessels, Vehicles and Machines	The Orient Express
Geographical locations	Switzerland
Social Organisations	IBM
Publications and works of the art	The Times
Languages and dialects	Hebrew

2.2.3 Pragmatics of Proper Names

Pragmatics, as defined in [18], is "The study or analysis of linguistic signs as they relate to the human user and his behaviour". Marmaridou (as seen in [19]) conducted an analysis of proper names from a pragmatic point of view. She claims that proper names are mainly exploited in two ways: *referential* and *connotative* [20].

The referential use of proper names is the function of identifying referents. For example, *Judas* in "Judas was Jesus Christ's disciple who betrayed him" is used to refer to the person who betrayed Jesus Christ.

In contrast, in the connotative use, proper names are metaphors used to offer shortcuts for ideas and thoughts, which are expressed implicitly. For instance, *Judas* in "Every great man nowadays has his disciples, but it is always Judas who writes the biography" does not specifically refer to anyone. It only refers to a kind of person who is likely to betray their master.

2.3 Computational Analysis of Proper Names

The computational analysis of proper names is non trivial for most natural language processing applications; Paik et al. shows its importance for information retrieval [22]. Radev and McKeown used a proper names extractor within a prototype system called PROFILE, which extracts noun-phrase descriptions of entities such as people, places and organisations [23]. Information Extraction (IE) and lexical knowledge acquisition are the two main areas where proper names analysis has been of great relevance.

2.3.1 Proper Names and Lexical Knowledge Acquisition

The analysis of proper names has become an essential area of computational linguistics in recent years. This has been evidenced by the fact that NLP systems created research laboratories to process large samples of unrestricted texts where proper names become noticeable, both in terms of ubiquity and complexity.

An accurate semantic and syntactic analysis of unrestricted texts requires a large lexicon. A simple approach to providing an NLP-style lexicon is to create Machine Readable

Dictionaries (MRDs). However, the fact that a great number of new words consistently arise from slang expressions, new coinages and especially from proper names makes it harder if not impossible for any lexicon aimed at naturally-occurring text to be completed [24]. Moreover, computational lexicographers do not consider entries for proper names in MRDs. They believe that such dictionaries should not include encyclopaedic entries (entries for proper names) because they are only slightly related to computational lexicography. Many researchers confirm the poor coverage of proper names in MRDs (see for example [25]). Therefore, robust NLP systems have to cope with unknown words like proper names.

Coates-Stephen [24] advocated the use of what he named as *within text descriptions* (WTDs) to tackle the poor lexical coverage of proper names. These WTDs include two kinds of syntactic patterns in which proper names are usually introduced in news articles. These patterns could comprise:

- A proper name followed by an apposition, which offers a description (e.g. Tony Blair, *The UK Prime Minister*).
- A descriptive word, which is either preceded or followed by a proper name (e.g. The American *President* George W. Bush, The *River* Nile).

This approach was implemented into an NLP system called FUNES. This system used a set of rules based on introspective knowledge of the appearance of proper names in news stories, to classify different classes of proper names including person, place, corporation and legislation names. An additional set of rules (heuristics) was exploited to make FUNES able to deal with proper names of more complicated structures like *University of Surrey* and *Victoria and Albert Museum*.

McDonald claims that the grammar of proper names is more lexical and less syntactic. Consequently, he invented two terms, *internal* and *external evidence* to point out both the structure of proper names and the context in which these names occur. McDonald's approach was considered as progress compared to earlier systems that had only relied on look up name lists to classify proper names [26].

Paik et *al.* analysed proper names for the benefit of information retrieval. They used proper names as important indications to retrieve relevant documents. This work used a categorisation and standardisation module to classify and convert variant forms of proper names into their canonical forms in order that different names for the same referent can be treated as one name. The proper name semantic categories used in this module are shown in Figure 2.2 below.

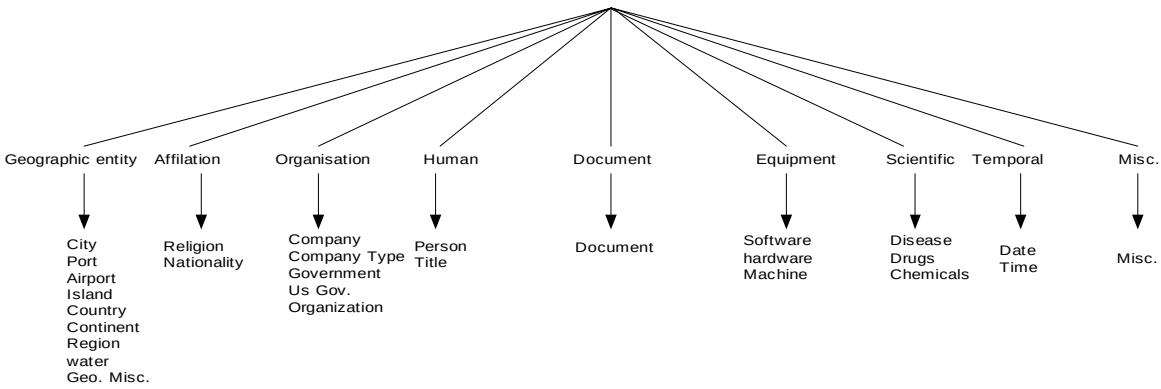

Figure 2.2: The semantic categories of proper names as defined by Paik et *al.* (1996).

2.3.2 Proper Names and IE

Nowadays, the Internet has made various sources of information available to the public, particularly textual documents. These documents are electronically accessible in different ways, for example, news wire, electronic journals, dictionaries and encyclopaedias. This has called for a serious change in the aims and strategies of NLP applications. Naturally occurring texts, including news articles, academic and scientific materials, become the key aim of most new NLP generations. These kinds of texts hold a significant amount of proper names.

IE, an NLP application, often takes short naturally occurring text as input, finds useful information in the text and converts it into structured format [21]. IE systems have mostly been concerned with factual and event-oriented texts (journalistic texts), which describe relationships between named entities. This speciality makes the extraction and understanding of proper names inevitable for IE.

Work on IE started in the 1960s (The Linguistic String Project at New York University) and grew very rapidly from 1980 when DARPA, the US defence agency, financed rival

research groups to pursue IE. Early IE systems tried to extract specific information from a variety of domains including medical, terrorism, satellite-flight information, social activities and biology. It was difficult to evaluate or even compare the performance of two different systems [19].

DARPA regularly sponsors so-called Message Understanding Conferences (MUCs) in order to establish a quantitative evaluation regime for IE systems. There have been seven MUCs to date, with the conference focus changing from one session to another. Table 2.3 below lists the domains that have been taken into consideration by MUCs so far.

Table 2.3: Message Understanding Conferences various domains.

MUC #	Text type	Domain	Date
1	Messages	Naval operations	1987
2			1989
3	News	Terrorist activities	1991
4			1992
5	News	Joint ventures and microelectronics	1993
6	News	Management successions	1995
7	News	Airplane Crashes, Space vehicles and missile launches	1997

Systems participating in the first five MUCs had to specify events in textual corpora of naval messages and news articles on terrorist events in Latin American countries as well as joint ventures and microelectronics. Therefore, there was an increasing need to accurately recognise named entities related to events of interest.

A separate subtask called *named-entity recognition* was especially designed for the extraction of proper names since MUC-6. This task is to recognise the following language expressions:

1. Proper names, including person, organisation and place names (ENAMEX).
2. Temporal expressions like dates and times (TIMEX).
3. Numerical expressions such as money and percentages (NUMEX).

Proper name analysis has largely gained from the developments in IE promoted by MUCs. More than thirty named entity extraction systems have been developed for

various languages (English, Spanish, Chinese, Japanese and Thai) with different approaches adopted in these systems:

Manually created rule-based systems: systems based on this approach use initially elaborated patterns that are applied to the text to identify included named entities. Here we mention the following:

NetOwl of IsoQuest (P&R = 91.32% for name tagging) consists of patterns and actions where patterns are similar to regular expressions and consist of special operators and operands that match portions of text. The action carries out certain operations on the matched text, such as classifying the extracted name [6]. A rule that extracts and classifies the proper name in phrases like *chief executive of Digital Equipment* as an organisation name is shown in Figure 2.3 below.

```
Pattern: job position + "of" + capitalised word sequence
Action: Tag match as ENTITY, excluding job position and "of"
```

Figure 2.3: A sample of the rules used with NetWol.

FACILE of UMIST (P&R = 79.5%), based on Coates-Stephens's approach in which he says "the text under analysis can act as a highly-tuned lexicon, in that it will describe many of the proper names it contains"[7], uses rules having the general form $A \Rightarrow B \setminus C / D$. Below is an example of these rules that captures University names [6].

```
[syn=NP , sem=ORG] (0.9) ⇒
\ [norm= "university"] , [token = "of"] , [sem=REGION|COUNTRY|CITY] / ;
```

Figure 2.4: A sample of the rules used with FACILE

FASTUS of SRI International Artificial Intelligence Centre is probably the most famous rule-based named entity extraction system. It uses a set of cascaded non-deterministic Finite State Automata (FSA), which helps increase its speed. Another interesting point with FASTUS is that users can write regular productions that are transferred into finite state machines using a special compiler. FASTUS achieved a 94 % F-measure in MUC-6.

Fully automated learning-based systems: these systems are generally based on statistics. They either use an *n*-gram language model (BBN's IdentiFinder and Kent Ridge Digital Labs) or maximum entropy models (MENE of New York University). Hence, the results can be a set of rules, a decision tree or a set of numeric data, which are difficult to revise, as the learning algorithm does not offer a readable output.

Hybrid approach: these systems are a combination of both rule-based and statistical systems. They automatically learn rules that are revised by experts or use an existing set of core rules (already defined by experts) that is dynamically modified so that better results are obtained. The systems of the University of Edinburgh (LTG), the University of Manitoba, and Cucchiarelli and Velardi are based on this approach [6]. FACILE of UMIST [7] also exploits this approach.

Table 2.4 below shows the name tagging evaluation results of both rule-based and machine-learning systems which participated in MUC-7. The results of the systems (F-score) are very close to the human experts (Annotators).

Table 2.4: MUC-7 Name Tagging Scores for English

Annotator	**F-Score** (P&R, β=1)[5]
Human Annotator 1*	99.03%
Human Annotator 2**	98.34%
HCRC-LTG	94.05%
IsoQuest	91.32%
BBN	90.61%
NYU	88.66%
U. of Manitoba	86.07%
U. of Sheffield	85.25%
MITRE	83.75%
Oki	81.13%
UMIST	79.50%
Kent Ridge	76.65%
U. of Durham	74.53%

However, these very good results are always obtained after manually tagging a very large number of words. For example, Nymble achieved a 90% success rate in MUC-6 at the cost of manually tagging around half a million words [5]. Moreover, these systems still exhibit drawbacks when applied to domains other than those considered in MUCs. This

[5] $$F = \frac{(\beta^2 + 1.0) \times P \times R}{(\beta^2 \times P) + R}$$

Where:
P is the precision.
R is the recall.
F is the weighted combination of recall and precision.
* Human annotation.
** Human annotation.

means while systems may perform well on the MUCs journalistic texts, they may not necessarily produce the same results when applied to different type of texts like e-mails and telephone transcripts [10]. The main explanation for this dramatic decline in performance was that sentences in these kinds of documents are ungrammatical and mal-formed. In addition, only certain named entity classes (person, organisation, place, dates, time and money) were considered of relevance. This categorisation policy may relieve the evaluation process, but it oversimplifies the categorisation of proper names [28]. This thesis, using the notion of *local grammar*, tries to reduce the high cost associated with rule compilation required when proper names are needed to be extracted from texts in new domains and languages.

2.4 Local Grammars

Harris (1991:272) defines a sublanguage as:

> ‘Certain proper subsets of the sentences of a language may be closed under some or all of the operations defined for the language. That is, carrying out a grammatical operation on a sentence of that subset yields another subset of the same subset. Such subsets constitute a sublanguage of the language, because they satisfy the grammar without involving any sentence that is outside the subset’.

Four types of sublanguages were suggested by Harris: grammar-based sublanguages – ‘composed of sentences which satisfy certain grammatical conditions which are not satisfied by all other sentences of the language’; the metalanguage - subset of the sentences of the object language which have their own grammar; subject-matter sublanguages – ‘composed of sentences which deal with more or less closed subject matter’; and science sublanguages (ibid: 273-278). Harris added that the frequent and specific use of word-combination in such subsets of languages turn out *special grammar* [29].

Sublanguages should all conform to or depart from characteristics such as:

(i) *Limited subject matter*

(ii) *Lexical, syntactic and semantic restrictions*

(iii) *Deviant grammar*

(iv) *High frequency of certain constructions*

(v) *Text structured and*

(vi) *Use of special symbols*

These characteristics are the analysis steps that should be followed in order to ascertain whether a language can be regarded as satisfying the requirement of the sublanguage model [30].

Hunston and Sinclair showed that there is an explicit link between the sublanguage and local grammar notions: 'It is possible, then, to see the items described by local grammars as small (but significant) sub-languages, and sub-language descriptions as extended local grammars' [31].

The local grammar movement was started by Harris (1991) to discuss recursive phrases that are commonly found in specialist literature like biochemistry (immunology) [29]. Gross (1993) extended the notion of local grammar to articulating dates and times [12]. Barnbrook (2002) suggested that dictionary definitions are formatted according to local grammar patterns [30]. Key-Sun and Jee-Sun (1997) applied much the same approach to Korean proper names [2]. MacDonald used it without mention to describe language construction containing proper names [26]. Shaikha Al-Jabir attempted to apply Sinclair and Barnbrook's work on dictionary definitions to those of terms in science and technology. She reported that around 80% of these term definitions are structurally similar to definitions found in the Cobuild dictionary [34]. Having observed many aspects of structural similarity with entries in general language and specialist dictionaries, Runli Guo suggested the extension of Sinclair's work on dictionary definitions to biographical entries [19]. The sections below describe their work in some details.

2.4.1 Language of Science (Biochemistry)

Harris analysed scientific articles on immunology in order to trace back statements that help clarify controversies in that line of research. He found few sentence types were used consistently to describe immunology material (see Table 2.5). Note that a subscript on X

means a subclass of X and a superscript on X means a modifier attached grammatically to X.

Table 2.5: Patterns in immunology reported by Harris.

Type[6]	Example
GJB	*Antigen is injected into a body part or an animal*
$GU^{ft}TT$	*Antigen moves from some tissue to some tissue*
TW and ***CW***	*A tissue (or cell) has some property or undergoes a change*
AVC	*Antibody appears in, is produced in, or is secreted from a cell*
CYC	*Some cell is similar to, or is called, some cell*
CY_cC	*A cell develops into another cell*

2.4.2 Date Expressions

Gross focused on how we specify times and dates and showed cardinal numbers, used to denote time and calendrical expressions (day/month/year, century), are embedded in their own local grammars [12]. He investigated *date phrases* that are preceded by the token *happen*. He initially found a set of repetitive date patterns used to denote dates rounded to the nearest ten. These patterns are semantically similar as any of them can be used to refer to an entire decade. Figure 2.5 below shows the reported set of these patterns represented in FSA.

[6] G: words for *antigen*'s synonyms or names.
A: words for *antibody*.
B: Bodies, body part, and animal names e.g. rabbit, footpad etc.
J: for *Inject*, which was used between *G and B*.
T: Tissue names.
C: Cell names.
V: class of verbs that occur between *A* and *C*.
W: words and adjectives occur after *T* or *C*.

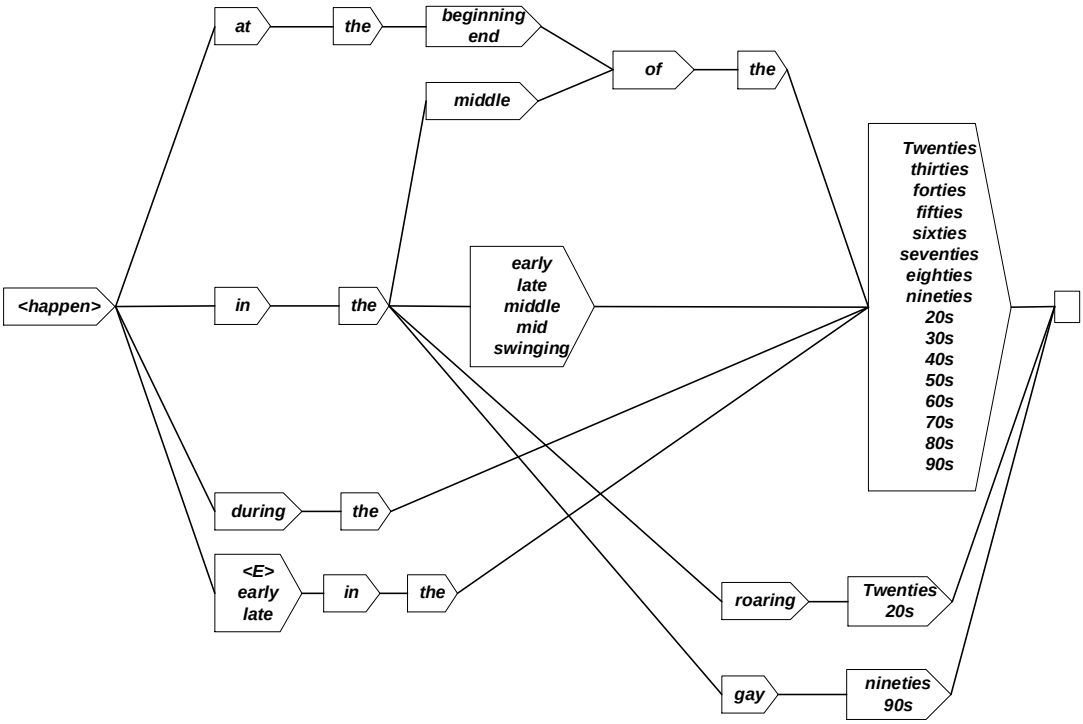

Figure 2.5: Local grammar for rounded dates (Gross, 1993).

Gross argues that more precise date phrases, e.g. *Tuesday September 7th, 1969*, also obey certain rules. These *rules* are stated as follows:

a. In the position of day name, any of the seven names of day may appear.

b. In the position of month name, any of the twelve names of month can occur.

c. In the position of numeral of days, we can find numerals from *1* to *31* with and without mark of order.

d. Finally, in the position of numeral of years a whole range of numerals is acceptable. However, numerals of years may have specific modifiers attached to them (e.g. 800 AD or 400 B.C.).

This date phrase, however, can be found in many other forms. Most of these forms are shown in Figure 2.6 below.

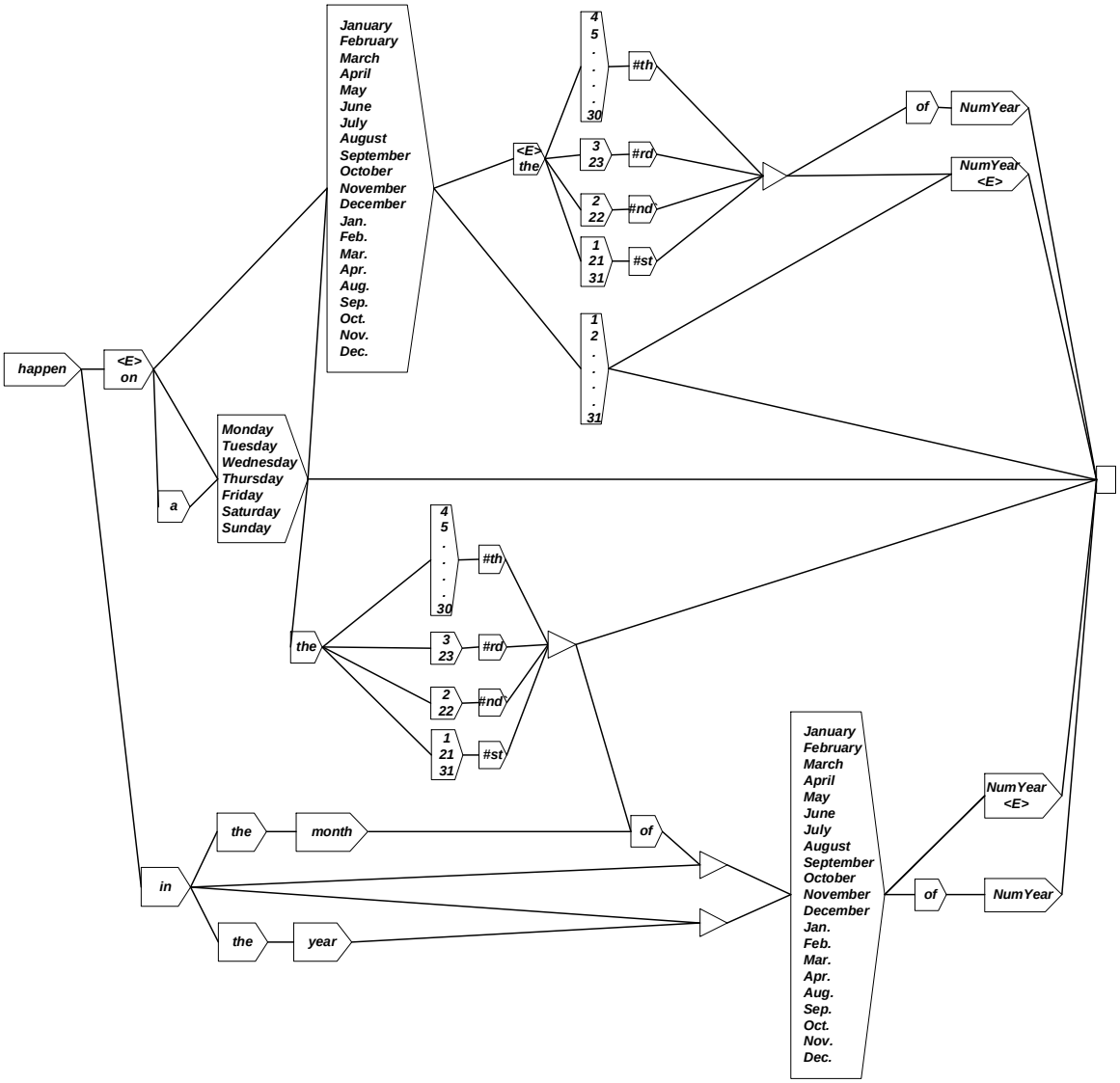

Figure 2.6: Local grammar for precise dates (Gross, 1993).

2.4.3 Dictionary Definitions - General Language

A definition is ‘A declaration or formal explanation of the signification of a word or phrase’ [18]. These declarations are usually written in clear and simple language, sometimes referred to as controlled *vocabulary*[7]. John Sinclair [32] argues that some

[7] If a *non-vocabulary* word occurs in a definition, a convention has been used in both Longman Dictionary of Contemporary English (LDOCE 1987) and Oxford Advanced Learner’s Dictionary (OALD 1995) to show such a word in ‘SMALL CAPITAL LETTERS’. However, when this word has different meanings, its sense number is also associated in order to direct the use to the right precise meaning. For example, the word atom in LDOC:

> *Atom: the smallest piece of simple substance (ELEMENT (6)) that still has the same qualities and can combine with other substances (to form MOLECULES).*

lexicographers formulate definitions for lexical items in 'distinctive and repetitive shapes according to their meanings'. Moreover, Sinclair described the language of definitions as ambiguity free (or can be rendered so) and structured according to a well-defined rule-base. Having analysed the definitions of *Collins Cobuild Student's Dictionary*, Barnbrook and Sinclair [33] reported the definition typology shown in Table 2.6.

Table 2.6: Types of Collins Cobuild Student's Dictionary defitions (Barnbrook and Sinclair, 1996).

Type	Explanation
A	*Definitions typically used for nouns and often begin with an article*
B	*Definitions typically define verbs and begin with* 'if' *or* 'when'
C	*Definitions used to mainly for verbs and begin with the infinitive marker* 'to' *and for adjectives, which have no text preceding the headword.*
D	*Definitions begin with* 'if' *or* 'when', *but are distinguished from type B by the nature of the remaining text before the headword which contains a phrase involving a reporting structure*
E	*Definitions begin with* 'something that' *or* 'someone who'
F	*Definitions begin with* 'if' *or* 'when' *and used for adjectives*
G	*Definitions begin with* 'if' *or* 'when' *and used for nouns*
H	*Definitions typically used for adjectives and often begin with an article*

In 1996, Barnbrook carried out a deeper analysis of *Cobuild Student's Dictionary* definitions. This analysis allowed him to organise these definitions into four major groups. Each group is subdivided into more fine grain syntactic patterns (types A to H). The first two groups contain over 52% of the total definitions and respectively deal with verbs and nouns. Table 2.7 illustrates the definition structures that are used in the three most frequent groups.

Table 2.7: General structure of definitions of types A to H together with the order of their occurrences in Cobuild.

Group	**Group1**			**Group2**			**Group3**	
Definition type	A	E	H	C	B	F	G	D
Initial article	1							
Cotext1	2	1			2	2	2	
Headword	3	3	2	2	3	4	5	3
Cotext2	4		3		4	5		4
Hinge	5		4	3		1	1	
Matching article	6							
Initial discriminator(s) and superordinate(s)	7							
Subsequent discriminator(s)	8							
Left-hand hinge		2				3	3	
Right-hand hinge		4				6	7	
Remaining definition text		5	5	4	6	8	8	
Projection								
Article			1				4	
Initial infinitive marker				1				1
Initial hinge					1			5
Matching pronouns					5			
Matching cotext1						7	6	
Frequency of occurrence	10494	2202	2212	1441	7528	1813	1714	561
General features	noun	default option after A,B,C,D,F,G	adj	Infinitive verbs and adj	Verb definition begin with when, if	adj	Noun definition begin with when, if	Verb definition begin with when, if

The results shown in the above table reveal a high frequency of a relatively small number of well-formed patterns that allowed the construction of a local grammar for each definition type. Figure 2.7[8] shows finite state automata representing the local grammar of definitions having the structure of types belong to Group1.

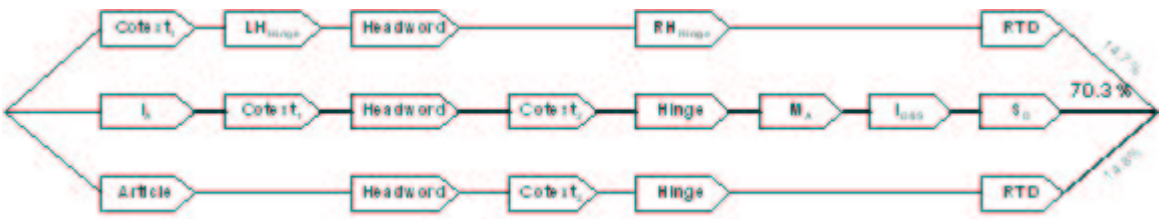

Figure 2.7: Local grammar for definitions of Group 1.

[8] LH: Left Hand
RH: Right Hand
RDT: Remaining Definition Text
I_A: Initial Article
$I_{D\&S}$: Initial Discriminator and superordinate.
S_D: Subsequent Discriminator.
M_A: Matching Article.

This grammar was used to develop a parser for dictionary definitions. This parser splits each definition into two parts, namely the left-part and the right-part. These parts map a headword to its meaning. Moreover, the left-part may contain a hinge (e.g. if, when) and cotexts. The right part generally holds a superordinate and at least one discriminator, a synonymic phrase or a synonym. The structures of both parts are shown in Figure 2.8 below.

Left-part: *[cotext 0] [hinge] [projection] [op-word] [context 1] [headword] [cotext 2] [hinge]*

Right-part: *[hinge] (superordinate discriminator) | synonymic phrase | synonym*

op-word: to, the, a, an.

cotext: words in the left-part which are not hinge or headword.

hinge: if, when, is, are, were, means, consists of.

projection: phrases involving reporting structure.

match: word or phrase contained in op-word and cotext, which may prospects a word or a phrase in the right-part.

Figure 2.8: Structure of left-part and right-part of Cobuild definitions, where items enclosed in [ ] are optional and | separates alternative items.

For example, if one of the definitions given to the word *abstract* in *Cobuild* dictionary is:

abstract [count n]

*An **abstract** of an article or speech is a short piece of writing that summarises the main point of it.*

The parser will return the following results:

grammar: count n

lemma: abstract abstracts abstracting abstracted

left-part: an abstract of an article or speech is

op-word(match_article(an)) + headword(abstract) + cotext2((of) + match(an article or speech)) + hinge(is)

right-part: a short piece of writing that summarises the main point of it

match_article(a) + discriminator(short) + superordinate(piece of writing) + discriminator(that summarises the main point of) + match1(it).

The results contain two different types of information, while the first type is purely lexical like *grammar* and *lemma*; the second is exceptionally related to the entries of the *Cobuild* dictionary. Such analysis is of great value, since it identifies the headword within a definition and relates it to its superordinate and discriminator elements. Hence, one can argue that this kind of analysis can be adapted to analyse phrases containing proper names and extract knowledge about them.

2.4.4 Term Entries – 'Special Language'

Terminology is a structured set of concepts. Every concept is represented by a lexical term and has a set of distinguishable characteristics. Terminologists compile these characteristics to devise the term definitions, which reflect the semantic specification contained in a concept and directed to human users with varying degrees of general knowledge.

Al-Jabir has analyzed definitions of terms in science and technology. Following Barnbrook and Sinclair, she carried out an initial analysis in which the frequency of the initial words of 1830 different terminology definitions has been reported. This analysis allowed the classification of these definitions according to their initial words and then comparison with *Cobuild* S*tudent's Dictionary* definitions. Table 2.8 below shows the results of this analysis [34].

Table 2.8: Frequency analysis of initial words of different terminology definitions [34].

Initial word of a definition	Term bases %	Cobuild student's dictionary %
a	29.9	21.6
the	22.4	4.7
no initial word	20.1	16.4
an	5.8	3.5
see	3.3	0
any	1.7	0
to	1.7	2.1
headword itself	1.3	0
one	1.1	0
in	1.0	0
this	0.7	0
all	0.6	0
when	0.5	4.7
having, two	0.4	0
also	0.3	0
if	0.3	32.4
something	0.2	3.3
very, referring, number, abbreviation, some, where	0.2	0.0
for, pertaining to, that, there, those, generally	0.1	0.0
someone	0.1	2.1
you	0.0	6.1
miscellaneous	0.1	0.0

Al-Jabir used these results to prove that more than 80% of terminology definitions begin with a noun phrase designating the superordinate. Hence the preponderance of a significant pattern of terminology definitions having the following structure:

superodinate phrase + discriminating phrase

Al-Jabir observed that the structure of this pattern could be further elaborated as:

[article][9] *[discriminator1] superodinate discriminator* *(Type A1)*

In addition, she noticed the absence of the *discriminator* element within some term definitions. Thus, the derivation of another type of pattern, which can be seen as:

synonymic phrase

or more precisely as:

[article] [discriminator1] synonym *(Type A2)*

Al-Jabir called these two dominant patterns type A1 and type A2 as they bear some similarity to type A definitions of Barnbrook and Sinclair. Examples of these types are illustrated in Table 2.9 below.

Table 2.9: Constructs of the initial phrases of the definitions with Type A1 or A2 [34].

Initial syntax of *Type A1* or *Type A2* definitions	**Example**
Headword + *is* + superordinate phrase	***Anaphylaxis*** Anaphylaxis is a medical emergency which involves an acute systemic (affecting the entire body) allergic reaction.
The plural form of the headword + *are* + superordinate phrase	***Sinus*** The sinuses are air cavities within the facial bones.
This + *is* + superordinate phrase	***ZE syndrome*** This is a disease in which abnormal amounts of a hormone called gastrin are produced.
These + *are* + superordinate phrase	***Liver function tests*** These are tests for the blood levels of bilirubin and also some enzymes normally present in liver cells.
Superordinate phrase	**Estrogen** A female hormone.

2.4.5 Biographical Entries

Having investigated the syntactic structure of biographical entries in American Heritage (1992), Larousse Dictionary of Scientists (1994), Oxford Dictionary of Music (1994) and Concise Oxford Dictionary of Linguistics (1997), Guo [19] has observed that biographers frequently write biographical entries using a template illustrated in Figure 2.9 below.

[9] Items enclosed between [ ] are optional

Headword:	The surname of a person
Hinge:	Year of birth [year of death]
Initial-discriminator(s):	Nationality [Nationality at birth (if different)]
Superordinate(s):	Profession [Other profession(s)]
Subsequent- discriminator:	Further attribute(s)

Figure 2.9: The structure of the template of biographical entries.

This template can be interpreted as follows:

Headword: the surname of a person, which is the key identifier of a person name.

Hinge: the set of digits connected by a hyphen and surrounded by a pair of brackets conveniently positioning themselves between the *headword* and the definition body.

definition body: the final part of a biographical entry. It provides information about the position or the status of the referent in relation to the rest of the world. Therefore, this part contains statements (like profession and nationality) to emphasise his or her significant contribution to the world.

Based on the fact that proper names are nouns, Guo argued that the above template is very similar to *Type A* as reported by Barnbrook and Sinclair in [30]:

Headword (*hinge*) [*initial-discriminator(s)*] *superordinate* [*subsequent-discriminator(s)*]

Accordingly, she classified biographical entries into five types:

Type A_1^{PN} : this type is the most frequent and simplest of the definitions of proper names. It does not have subsequent-discriminators. The biographical entry below is an example:

Feynman, Richard Philips (1918-88), American theoretical physicist.

Type A_2^{PN} : this type is a variant of Type A_1^{PN} . It has more than one superordinate. Consider the following entry of the Finnish architect and designer Alvar Aalto.

Aalto, Alvar (1898-1976), Finnish architect and designer.

Type A_3^{PN} : this type was reported to be less frequent than the previous types. It has initial and subsequent discriminators. The subsequent discriminator could be an adjectival phrase or adjectival clause. The two entries below describe this type.

Einstein, Albert (1879-1955), German-American physicist who …[10]

Marc, Franz (1880-1916), German expressionist painter associated with the Blue Reiter group of artists.

Figure 2.10 shows the local grammar for the above types of biographical entries reported by Runli Guo.

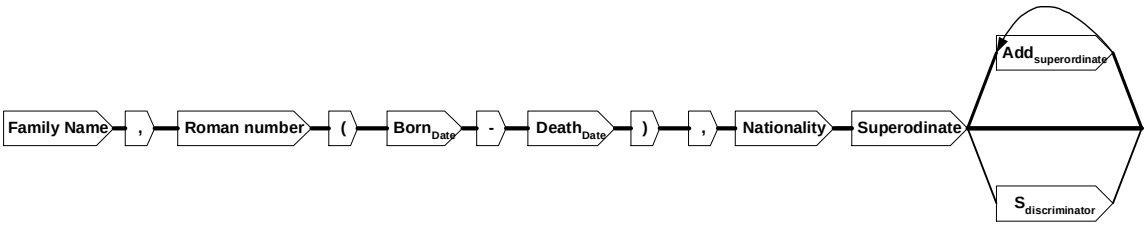

Figure 2.10: FSA cascade formalising frequent structures of biographical entries.

Type A_4^{PN} : this type was specifically found in the *Pocket Oxford Dictionary of Biography* (1997). The superordinate was reported to be absent in this type of biographical entry. Consider the following entry:

Elizabeth, I (1533-1601), daughter of Henry VIII.

The *definition body* of this type is an opposition that expresses the relationship between the *headword* and another named entity that is equally or even better known. This type was observed to occur frequently for a person who plays an important role within a specific organisation. Semantically, this type has the following structure:

Headword (hinge) affiliation affiliate

Complex types: this expression was given to entries that contain all attributes of *Type A* of Banbrook and Sinclair. Consider the following example:

Lie, Trygve Halvdan (1896-1968), Norwegian Labour politician, first Secretary-General of the United Nations 1946-53.

This analysis revealed a set of repetitive patterns used in biographies. Guo suggested the use of these patterns to map biographical dictionaries onto a knowledge base, which can be used to provide world knowledge of person names likely to occur in free texts like newspapers, magazines and other documents. Biographical dictionaries, in our opinion, would not be adequate enough for dealing with news articles where the biography of most

[10] http://scienceworld.wolfram.com/biography/Einstein.html

included person names has not been written as yet. The next chapter shows how a reasonable amount of world knowledge about person names can be achieved from Reuters' financial news articles.

2.4.6 Proper Names and Local Grammar

Kee-Sun Choi et *al.* used the notion of local grammar to extract proper names from Korean texts [2]. They adopted this approach after having realised that general syntactic rules (generative grammar), apart from lexical items, cannot provide good analyses. Alternatively, they advocate the investigation of the syntactic contexts of some lexical items (e.g. professional titles, or family relations) where proper names would appear in Korean texts. This strategy allowed the construction of five context types (or local grammars) describing proper name occurrences. The structures of these contexts are shown in Table 2.10.

Table 2.10: Context types for Korean proper names.

Type	Structure	Example
Type I	PN- (Postposition + E[11])	Kim Jung II is the president of North Korea
Type II	PN- (Spec+ E)–Professional Title– (Postposition+E)	Kim Minu Dr. *has studied in USA during 5 years*
Type III	PN-(Gen + E)-Family Relation-(Postposition+E)	Kim Minu's son is 20 yearsold this year
Type IV	PN-Vocative Term-(Postposition+E)	Kim Sir came
Type V	PN Incomplete Noun-(Postposition+E)	Kim Mr. Minu came

Type I considers proper names preceding postpositions. Proper names appearing in such context are difficult to distinguish from common nouns using only capitalisation rules as we can also say (This man + A Korean) is the president of North Korea. Figure 2.11 below shows the FSA that represents this type.

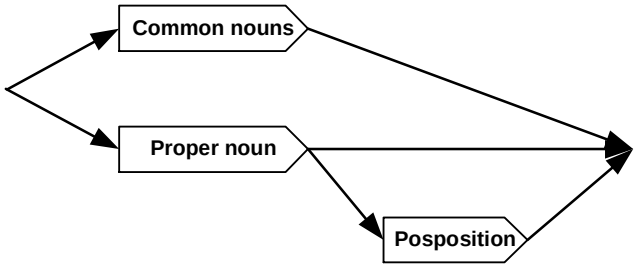

Figure 2.11: FSA for Type I.

[11] E means this part can be empty.

Type II is characterised by the presence of a professional title, e.g. Doctor, Professor etc., straight after the proper name. This type gives the option for words (W) designating speciality domains such 'physics', 'biology' or 'mathematics' for proper names to occur before the professional title. The FSA formalising this type is illustrated in Figure 2.12 below.

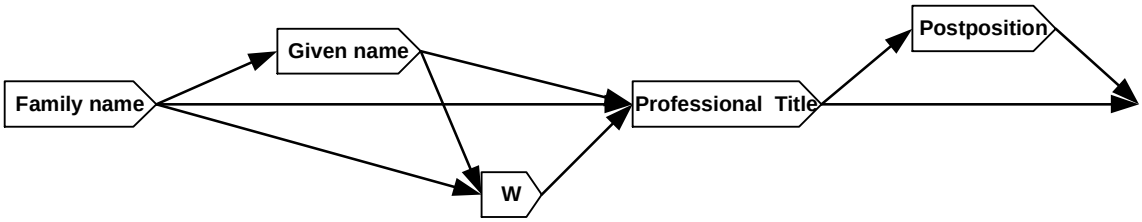

Figure 2.12: FSA for Type II.

Type III contains nouns indicating a family relation such as son, father, etc. Kee-Sun Choi et al. argued that these nouns frequently occur in the neighbourhood of proper names. They also reported other variants of this type which may contain a genitive postposition (Gen) as shown in Figure 2.13 below.

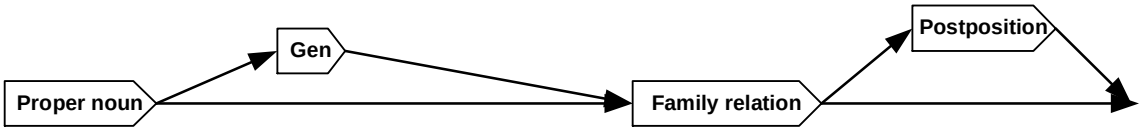

Figure 2.13: FSA for Type III.

Type IV and V differ only in the type of nouns used for social appellation. The former contains what the authors called *vocative terms* such as Sir, Senior, Elder sister or Elder brother. The latter holds what they called *incomplete nouns* like Mr or Miss.

Eventually, this local grammar was compiled on the basis of introspective knowledge of the Korean language. Kee-Sun Choi *et al*. did not mention any method used to define the lexical items they used as clues to the presence of proper names. In addition, the local grammar they suggested would only recognise proper names out of DECOS (a lexicon of Korean common nouns). Therefore, creative proper names (names borrowed from common nouns) cannot be recognised. The main aim of their work was to recognise proper names automatically on a large scale in texts in order to construct an electronic lexicon of Korean proper names.

Friburger *et al.* [11] suggested the use of finite state automata and particularly transducers to recognise proper names in French journalistic texts. Their work forms a part of the PROLEX project, based at the Université de Tours. This project, sponsored by the French Ministry of Research's Department of Information and Communication Technologies, aims at the automatic treatment of proper names. Friburger *et al*. used the INTEX[12] system, which is a linguistic development environment that includes large-coverage dictionaries and grammars, to pre-process French texts. The pre-processing stage includes splitting texts into sentences and then tagging them with place name, first name and acronym dictionaries. Then finite state transducer cascades are applied to the text in order to extract included proper names. The transducers are constructed after investigating the right and left contexts of person names contained in a corpus of Le Monde newspaper (about 165000 words). Table 2.11 below shows the frequency distribution of reported contexts (patterns).

Table 2.11: The frequency distribution of contexts of person names reported by Friburger *et al*.

Case	Freq	Context	Example
1	29.9%	Person names (full names) appear after a title or an occupation name	*Le presidentJohn Kenned*: **President John Kennedy**
2	19.1%	Person names (only patronymic names) occur after a title or an occupation name	*Le president Chadli* : **President Chadli**
3	43.4%	Person names (full names) occur without any accompanied descriptions	*Pierre Bourdieu*
4	5.2%	Person names located after verbs denoting human actions such as *to say* or *to explain*. This category includes appositions as well.	*Wieviorka est decidé* : **Wieviorka decided**
			Jospin, Premier Minister : **Jospin, Prime Minister**
5	6.4%	Person names not associated with any description (first name, title or profession) that can be used to differentiate them from other proper names. 49 % of these names belong to famous people.	*Bush, Blair, Chirac* etc.

These patterns can only deal with the extraction of person names. Their compilation was very costly – the left and right contexts of person names were manually checked. In addition, such patterns are merely able to define the semantic category (person names) of involved proper names – this can only help the syntactic analysis of a given text, but not the semantic understanding of it. The understanding process requires more world

[12] INTEX. Available at: http://www.nyu.edu/pages/linguistics/intex/. Last visited on 28/01/04.

knowledge about person names to be provided [19]. For example, the identification of Jospin as a French Prime Minister would tell the reader which prime minister the article talks about. The local grammar reported in this thesis enables the acquisition of information (if available) from Reuters' news items, which helps in referring a person to the organisation he belongs to.

2.5 Conclusion

In this chapter we have tried to understand proper names from onomastic and linguistic perspectives. While the onomastic perspective helped us to know more about the nature (origins and forms) of proper names, the linguistic studies gave us some guideline to improve our knowledge about their syntactic, semantic and pragmatic characteristics. More importantly, we have explored approaches adopted for automatic analysis of proper names. This review has revealed some outstanding issues: (a) work in automatic analysis of proper names has been mostly restricted to news-wire genres; (b) The manual construction of rule bases, integrated in NE extraction systems using the linguistic approach, is introduced as a costly and lengthy process; and, (c) NE recognition tools developed for news articles are generally not sufficient to deal with texts in other domains unless the context of proper names in such domains is thoroughly revised. We have also looked into studies aimed at ungrammatical ways of describing the local grammar of language expressions including dates, Korean proper names and definitions in both general and special language dictionaries and suggested that the same approach can be introduced to deal with proper names in news article texts.

The following chapter studies the language used in news articles in order to extract knowledge about proper names. An automatic corpus-based method is used to identify patterns used by journalists to introduce person and organisation names. The local grammar which governs these patterns is examined for the automatic extraction of these names.

Chapter 3

3 A Local Grammar for Proper Names

3.1 Introduction

The analysis of proper names is of great significance to IE and text understanding. Researchers have adopted both linguistic and statistical approaches to tackle this task. Systems based on the linguistic approach use bases of rules describing consistently used contexts of proper names at both the syntactic and lexical levels. These rules are manually compiled and need to be revised every time proper names are to be identified in different kinds of documents. The manual compilation of rules that help the recognition of proper names in texts is known to be a lengthy and boring process [11]. This chapter shows a method to automatically identify patterns of proper names in a specific domain by first defining keywords denoting human actions in the domain and then applying techniques in corpus linguistics including frequency, collocation and concordance analyses, to study the structure of contexts containing proper names and keywords. So far this method has been applied only to a corpus of financial news articles; wider applications of this method are planned for future work.

3.2 A Method

Documents written in English comprise two major linguistic units: the highly frequent closed class words, determiners, prepositions, conjunctions and some verbs; and lower frequency open class words such as nouns, adjectives, adverbs and modal verbs used in technical or specialist languages. In general, closed class words are indicative of the natural language being used and open class words indicates the topic of discussion. In everyday language, fewer open class words are used than in special language texts. Different specialists use distinct open class words with much higher frequency.

Local grammars are rules that govern the simultaneous choice of a set of words used in a specialist context, for example, sentences/clauses used for telling time and dates. Others include what appear to be telegraphic statements in specialist domains (see Table 3.1).

Table 3.1: Local grammars in different specialistt domains.

Structure	Source
Molecule + Process + Solvent	**Harris 1991**
The polypeptides were washed in acid *The protein was treated with acid.*	
Accidents + Transport + Numbers	**BNC Online**
accidents including cars rose/fell from # to # *accidents including ships rose/fell from # to #*	
Instrument + Movement + Numbers	**Gross 1997**
The Dow Jones fell 14.50 points. *Dow Jones rose 15 points.*	

A local grammar may sometimes violate the universal grammar model but works well in its own domain. Gross (1997) has discussed local grammars for the financial domain to suggest some local grammars for sentences that are used to indicate the status of a financial instrument [36]. But how do we define the local grammars? Our work is an extension of Gross in that we suggest that a corpus of texts can be used systematically to find such grammars. The local grammars deal with a large number of open class words – nouns, verbs, adjective and restricted closed class words such prepositions, numbers and punctuation marks. The key point, at least for us, is that open class words are frequently used in specialist domains. And it is this simultaneous use of frequent open class words with other well-defined words which might lead to meaning-bearing sentences that are governed by a local grammar. Typically, such meaning-bearing sentences are attributed to people or organisations of reputation and authority.

A related notion to that of local grammar is that of collocation. Collocation is related to the localist *idiom principle*, as opposed to the *open choice principle* governed by (universal) grammaticalness [39]. There are frequently used words and less frequent words that are simultaneously used. For example, the frequently used “washed” may occur with many proteins and acids in the example above. Many financial instruments, currencies, indices, co-occur with the frequently used *fell* and *rose*. In addition *said*, *told* can be used with a number of people and organization names. This, collocation between high frequency words and many lower frequency words is called “downward collocation” [39].

We present a systematic method of finding such collocation between the frequently occurring words in a text corpus and other lower/higher frequency words. The collocation between non-neighbouring collocates may be just as useful between neighbouring collocates. The strength of their collocations can be statistically tested under certain statistical assumptions [37].

Starting from a corpus of specialist text, most frequently used words are selected and their collocations are identified. A set of concordances is produced and visually examined. This examination leads to a local grammar. The local grammar is then used either for building a linguistic description of the language governing specialist texts or for purposes of recognising patterns of local grammar usage in unseen texts. Such recognition is useful for information extraction tasks. Furthermore, using the above procedure for identifying such patterns and adding them to the local grammar can incorporate unrecognised patterns. This updating of local grammars using unseen texts is an example of adaptation or perhaps more formally that of adaptive learning (see Figure 3.1).

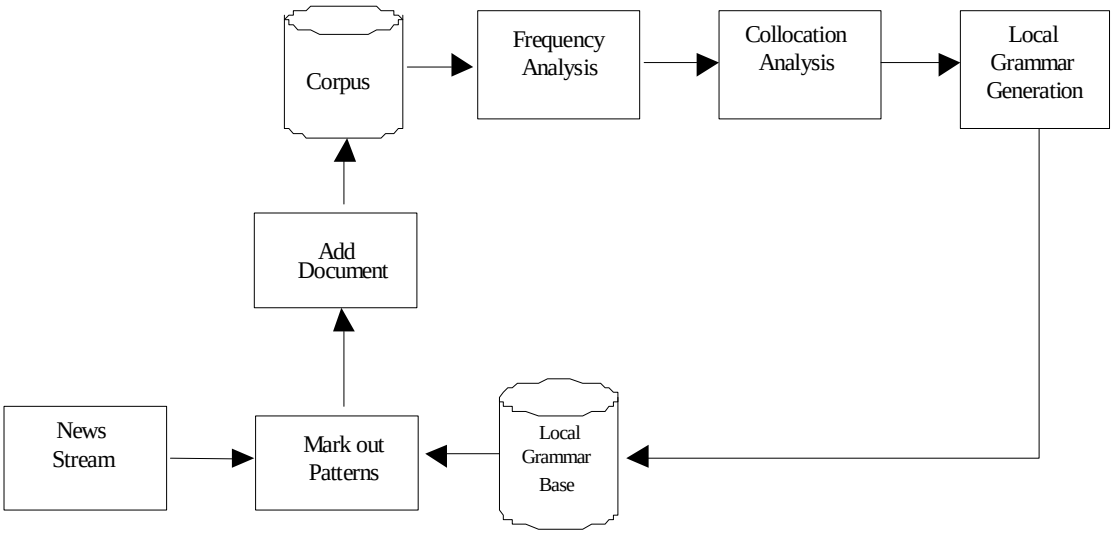

Figure 3.1: Analysis stages to discover patterns.

3.3 Data

We have used Reuters Financial News Service as the basis to build our corpus. Our Reuters corpus comprises 9,063 XML-formatted texts comprising 3.63 million tokens

published during Jan – Dec 2002. The average length of the news reports is 400 tokens. The subjects typically covered by Reuters Financial News Service are: Company Outlooks, Company Results, Economic Indicators, Funds and Initial Public Offering News.

3.4 Analyses

3.4.1 Frequency Analysis

A frequency analysis of the corpus shows that the frequency of the first 50 most frequent tokens accounts for 40% (1.45 million tokens). The first 10 most frequent tokens includes, in addition to some closed class words, key open class words like *said* (43643) and *percent* (41083) (see Table 3.2).

Table 3.2: Distribution of the first 100 most frequent tokens in both Reuters training and testing financial corpora according to the cumulative frequency of ten tokens at a time.

Rank	Token	Cumulative Relative Frequency
1-10	the, to, of, in, a, and, *s*, **said**, on, **percent**	21.70%
11-20	for, it, its, that, by, at, was, is, as, with	7.30%
21-30	from, **year**, has, but, be, **million**, which, after, have, **market**	4.44%
31-40	would, up, **pounds, shares**, will, are, **Reuters**, had, this, an	3.37%
41-50	were, **London**, not, u, he, we, **company, group**, last, **billion**	2.71%
51-60	**bank**, ***more***, than, ***over***, also, one, been, ***new***, down, about	2.08%
61-70	**FTSE**, they, **sales, index**, ***first***, **firm**, **bst, growth**, their, some	1.81%
71-80	or, **business**, there, ***two***, **points**, could, expected, **pence, quarter, UK**	1.55%
81-90	out, **share, stock, sector, financial, investors, profits, analysts**, off, ***three***	1.32%
91-100	while, **price**, since, **may**, ***half***, **month, prices, months**, if, into	1.19%
	TOTAL	**47.47%**

The high frequency of the token *said*, which is usually used to illustrate a person's speech in narrative texts, was the main motivation for further investigation into its contextual use.

3.4.2 Collocation Analysis

Here we follow a method outlined by Smadja for retrieving collocations from text [37]. Generally, collocations are word pairs. The pair may be neighbours, or may co-occur with other interspersing words: up to five left neighbours and five right neighbours of a high frequency word appear to be significant. Smadja defines a peak or lexical relation

containing a high frequency word *w* as a tuple (w_i, *distance*, *strength*, *spread*, *j*) that verifies the following inqualities:

$$strength = \frac{freq_i - \overline{f}}{\sigma} \geq k_0 \qquad \text{(a)}$$

$$spread \geq U_0 \qquad \text{(b)}$$

$$p_j^i \geq \overline{p_i} + (k_1 \times \sqrt{U_i}) \qquad \text{(c)}$$

Where

(a) is used to eliminate low frequency collocates. The $freq_i$ is the frequency of the collocation of w_i with *w*; $\overline{f}$ is the average frequency, σ is the standard deviation and k_0 is the strength threshold. This threshold usually has a value of one for the task of language generation [38].

(b) requires that the histogram of the ten relative frequencies of appearance of w_i within five words of *w* to have at least one spike. The histograms are rejected if the variance threshold $U_0 < 10$. The variance is usually computed using the equation below.

$$U_i = \frac{\sum_{j=1}^{10} \left(p_i^j - \overline{p_i} \right)^2}{10}$$

where p_j^i (*j* in the above tuple) and $\overline{p}_i$ are the frequency of one collocate at certain distance from *w* and their average respectively.

(c) pulls out the significant relative positions of two words. Thus, this inequality eliminates columns whereas (a) and (b) select rows. It states that the frequency threshold of one collocate at certain distance from *w* has to be at least one standard deviation above the average frequency of one row collocates ($k_1 = 1$). However, the value of K_1 is task dependent and can be adjusted to suit the task in hand [38]. We found that $k_1 = 0.65$ gives good results for texts of news articles.

Looking at the most frequent collocates of *said* in Table 3.3 we see that the definite article *the* occurred 203386 times in the whole corpus, of which, 17210 instances are distributed in ten different positions before and after the token *said*. Whereas, the punctuation mark (“) occurred 60148 times, of which, 13542 clustered to the left of *said* and only 22 instances appeared to its right.

Table 3.3: Collocates of the *said* (f=43416, $\overline{f}$ = 19.48, σ = 230.80, k_0 = 1, U_0 = 10 and k_1 = 0.65).

		$f_{neighbourhood\ of\ said}$													
	f	**-5**	**-4**	**-3**	**-2**	**-1**	**1**	**2**	**3**	**4**	**5**	***f***	**K-score**	**U-score**	**$Freq_{Th}$**
Said	43416														
The	203386	1543	1834	1678	**2579**	2	**5005**	672	1257	1242	1398	17210	74.47	1617155	2548
"	60148	452	424	1308	**4329**	**7007**	5	3	4	6	4	13542	58.58	5159510	2831
A	82187	689	726	501	152	2	872	**2193**	694	836	742	7407	32	309027	1102
analyst	2542	55	104	231	10	198	82	221	**442**	**482**	145	1970	8.45	22269	294
After	14663	198	180	242	**693**	1	79	30	20	28	42	1513	6.47	39159	280
Chief	4013	117	**514**	28	42	4	117	83	**289**	49	25	1268	5.40	22679	225
But	15881	74	79	120	**284**	**331**	24	1	2	11	33	959	4.07	12615	169
analysts	4611	18	31	39	40	**619**	42	13	14	13	31	860	3.64	39512	215

The distribution of the indefinite article *a* was almost similar to the definite one – on both sides of *said*. The other collocates were ignored because their *K-score* and *U-score* were less than the standards. The results we obtained from the collocation analysis can be interpreted as follows:

- The appearance of the definite article in the neighbourhood of *said*, especially to its left, raise the chance of finding proper names near *said*. This type of article is frequently used with proper names to reflect their uniqueness (e.g. *the BBC said*).
- The frequent occurrence of the quotation mark (“) to the left of *said* indicates the existence of someone’s quoted speech before *said*.
- The indefinite article *a* normally does not fix the identity of the noun modified. Journalists (if asked) do not name their sources (e.g *A Treasury spokesman said*).

The results shown in the collocation table above reveal a high frequency of a relatively small number of well-formed patterns containing *said*. However, their structures still need to be investigated.

3.4.3 Concordance Analysis

The collocation analysis gave us some general clues of the external structure of some frequent patterns used to introduce the speech of market players (see Table 3.4 below). Therefore, a concordance analysis for each of these patterns was needed to elaborate their internal structures.

Table 3.4: *said* most frequent patterns.

Pattern	Frequency	% Frequency
the […] ***said***	8457	18 %
“ said […] .	7733	17%
“ […] ***said***	7257	15.6%
a […] ***said***	2310	5 %

Although, these patterns could be exploited to introduce the speech of anonymous entities (e.g. *the company said*, *a bank spokeswoman said*, *"he said*, and *"said a dealer*), the concordance analysis demonstrated that the occurrence of expressions containing proper names within these patterns is definite. We will refer to these expressions as *NE*. Examples of concordances containing *NE* expressions are depicted in Figure 3.2.

“[…]” said Thomas Deitz, media analyst at Merrill Lynch .

“[…]” Princess Chief Executive Peter Ratcliffe ***said***

The Federal Reserve Bank of Philadelphia **said**

a YES spokesman **said**

Figure 3.2: Concordances of the most frequent patterns.

Since this thesis is concerned with the extraction of proper names, we will only study the structure of patterns containing *NE* expressions. Thus, patterns introduced in Table 3.4 become: ***“ said*** *NE .*, ***“*** *NE* ***said, the*** *NE* ***said***, ***a*** *NE* ***said.*** Table 3.5 illustrates the frequency distribution of each of these patterns across the whole corpus.

Table 3.5: The frequency distribution of patterns containing proper names across the whole corpus.

Collocate	Testing Corpus										Training Corpus		Avg	STD	Freq	Z-score
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Dec	Oct	Nov				
f_{said}	3817	3712	2891	3193	3080	3075	4611	2983	3864	3203	4584	4630	3637	668.57	43643	59.84
$f_{said + collocate}$	1796	1650	1256	1492	1520	1544	2381	1521	1756	1564	2065	1979	1710	306.90	20524	61.30
“[…]” said NE.	466	438	303	352	472	443	680	440	529	435	558	623	478	106.01	5739	49.62
“[…]” NE said	310	251	219	271	269	266	453	283	223	244	265	386	287	68.27	3440	46.19
the NE said	162	118	71	102	87	97	139	123	103	103	132	131	114	25.03	1368	50.10
a NE said	28	20	21	41	10	17	22	20	25	26	23	23	23	7.32	276	34.55

This analysis shows that *said* occurred very frequently in contexts containing proper names (47%). Around 53% of these contexts contain *NE* expressions. These expressions define the originator of one speech. They mostly comprise two parts:

1. The *name* part: this part contains the speech originator name.
2. The *defining* part: this part includes world knowledge of the speech originator. It provides information about the position or the status of the referent in relation to the rest of the world.

However, there are cases where the *defining* part tends to be missing or exists on its own. These cases occurr commonly after definite and indefinite articles. Consider, for example, ***the*** *Regions Select Committee* ***said***, ***"[…]",*** *Bear Stearns* ***said*** and ***a*** *Corus spokesman* ***said***. These *NE* expressions appear to have many variants. The structures of most frequent *NE* variants are shown in Table 3.6 below.

Table 3.6: The structure of the most frequent of *NE* variants.

Pattern	Variant♣	% Frequency
***"[…]" said** NE.*	***"[…]" said** PN, TITLE at ORG.*	23%
	***"[…]" said** ORG TITLE PN.*	6%
	***"[…]" said** ORG/PN.*	5.4%
	***"[…]" said** PN, TITLE at MOD ORG.*	4.3%
	***"[…]" said** PN at ORG.*	3.7%
	***"[…]" said** PN, who*	2.8%
	***"[…]" said** PN, TITLE at ORG in PLACE.*	2.1%
	***"[…]" said** PN of ORG.*	1.9%
	***"[…]" said** PN at MOD ORG.*	1.8%
	***"[…]" said** TITLE PN.*	1.7%
	***"[…]" said** PN, an TITLE at ORG.*	1.5%
	***"[…]" said** ORG's PN.*	1.4%
	***"[…]" said** ORG TITLE PN, who*	1.1%
"[…]"** NE **said	*"**[…]**" ORG/PN **said***	58%
	*"**[…]**" TITLE PN **said***	10.7%
	*"**[…]**" ORG TITLE PN **said***	9.7 %
	*"**[…]**" the ORG **said***	2.4%
	*"**[…]**" a ORG TITLE **said***	2.2%
	*"**[…]**"ORG's PN **said***	1.7%
	*"**[…]**" a TITLE for ORG **said***	0.9%
the** NE **said	***the** ORG **said***	100%
a** NE **said	***a** ORG TITLE **said***	46%
	a** TITLE for ORG **said	13.4%
	a** TITLE for the ORG **said	5.8%
	a** source close to ORG **said	2.5%
	a** ORG TITLE in PLACE **said	2.2%
	a** ORG source **said	1.8%
	a** survey by the ORG **said	1.4%
	a** TITLE for ORG on DATE **said	1.4%
	a** TITLE at ORG **said	1.1%
	a** TITLE for ORG in PLACE **said	1.1%

The results revealed in Table 3.6 show that there are some dominant variants of *NE* expressions embedded in *said* patterns. The domination of one pattern form over others with related meanings helped isolate and build the local grammar of four sets of *said* patterns. We used Finite State Automata (FSA) to represent their local grammars.

Figure 3.3 shows a cascade of FSA indicating variants of ***"[…]" said** NE.* The most frequent amongst these variants contain a *NE* expression having:

♣ *PN*: person name.
ORG: organisation name.
MOD: Modifier used before the organisation name (e.g. banker).
ORG/PN: used when the category of the involved name is not known (e.g. *"[…]" said Young.*).
TITLE: the professional title of the speech originator.

1. *Name* and *defining* parts separated by a comma - *PN, TITLE at ORG*. The name part contains a person name (*PN*) and the *defining* part comprises his or her professional title (*TITLE*) and the name of the organisation (*ORG*) they are employed by (e.g. ***"[…]" said*** *A.C. Moore, chief investment strategist at Dunvegan Associates.*). Having assumed that the comma plays the *Hinge* role, This *NE* becomes similar to Type A_4^{PN} defined by Guo: *Headword (hinge) affiliation affiliate*.
2. *Name* and *defining* parts without any separator - *ORG TITLE PN*. The *defining* part in this *NE* precedes the *name* part (***"[…]" said*** *Bank of America analyst Jonathan Gollins.*).
3. *Name* part only – *ORG/PN*. The *defining* part is absent in this type of *NE* (e.g. ***"[…]" said*** *Branson.*).

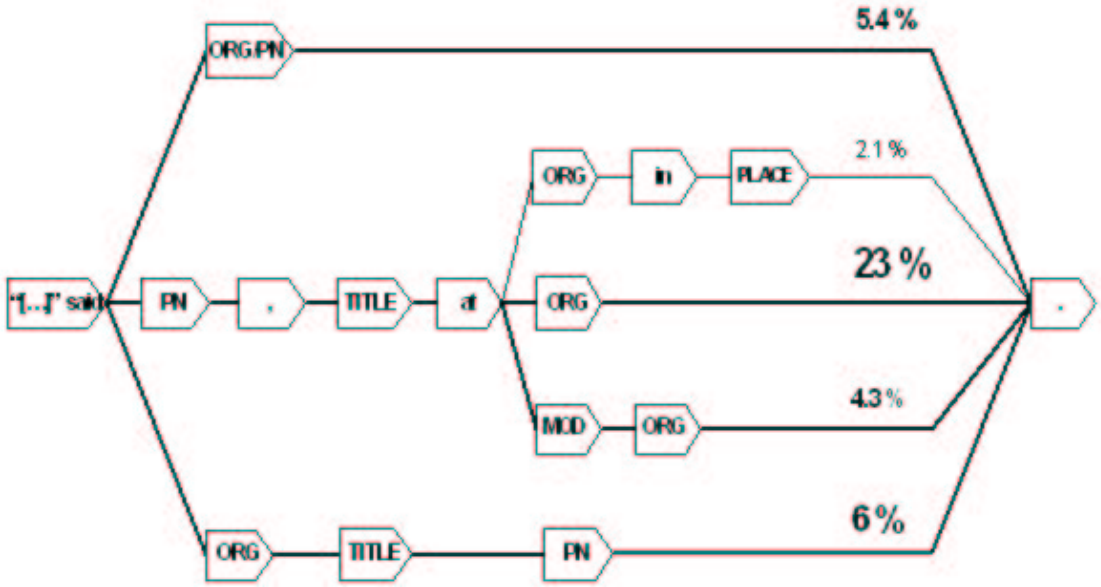

Figure 3.3: Cascade of FSA representing some of the most frequent variants of *"[…]" said NE.*

Figure 3.4 details FSA of ***"[…]"*** *NE* ***said*** variants. The concordance analysis showed that:

- 58% of these variants enclose a *NE* containing only a *name* part of one word length (e.g. ***"[…]"*** *Davies* ***said***). This *NE* refers to either a person surname or organisation name.
- 10.7% have a *defining* part (*TITLE*) followed by a *name* part containing a person name (e.g. *"[…]" analyst Iain McDonald said*).
- 9.7% comprise a *name* part preceded by a *defining* part (e.g. *"[…]" BNP Paribas analyst Conor O'Shea said*).

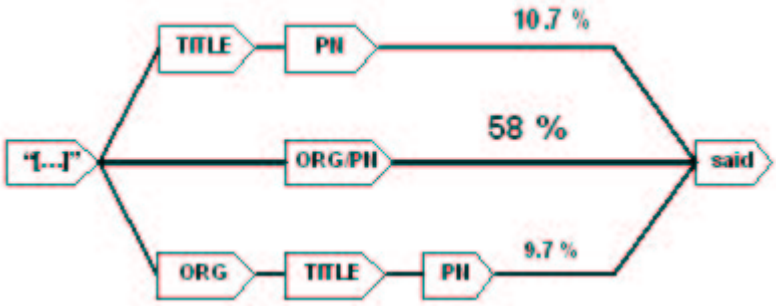

Figure 3.4: Cascade of FSA representing the most frequent variants of *"[…]" NE said.*

Figure 3.5 illustrates the most frequent variant of ***the*** *NE* ***said***. The *NE* in this *said* pattern has only a name part. The fact that the definite article is not used with person names caused the name part in this pattern to contain only the organisation name (*ORG*). This perhaps justifies the absence of the defining part.

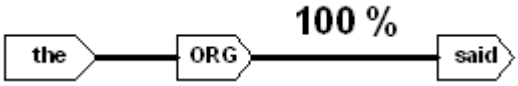

Figure 3.5: FSA representing the most frequent variant of *the NE said.*

Finally, Figure 3.6 describes FSA depicting variants of the less frequent *said* pattern - ***a*** *NE* ***said.*** This pattern has three main variants. The name part is found to be missing in each of the *NE* expressions of these variants. The indefinite article does not fix the identity of the noun modified (e.g. ***a*** *Boeing spokesman* ***said***).

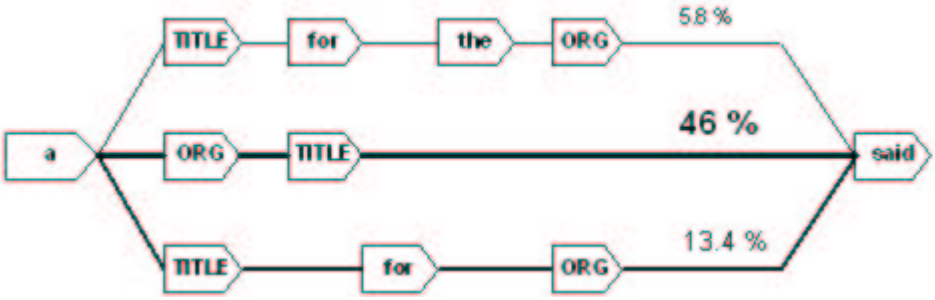

Figure 3.6: Cascade of FSA representing *a NE said.*

3.5 Conclusion

Throughout this chapter we have discussed the theoretical framework on which our work is based. The framework is based on our observation of a sublanguage in news stories used to introduce market player statements. Our analysis of the context of these

statements leads to the conclusion that there are recursive patterns which are always exploited in English news items.

Typically, paraphrased and quoted statements happen in the neighbourhood of the word *said*. The context of these statements is often associated with expressions containing proper names. We called these expressions *NE*. The analysis of these expressions showed that they mostly contain two parts: name and defining parts. The name part comprises the name of the statement originator (person or organisation) whereas the defining part encloses world knowledge of the statement originator.

Based on these findings we have built an archetype which extracts person and organisation names along with information that might connect these names. This system is named NExtract. In the following chapter, we will discuss all the aspects of this archetype including the specifications, design, implementation as well as the evaluation.

Chapter 4

4 An Archetype for Proper Names Analysis

A natural language is a language spoken or written by humans which is characterised by a lack of structure and is unlike languages used to program or communicate with computers. The complexity, irregularity and diversity of human language and the philosophical problems of meaning make the processing and understanding of such language amongst the hardest problems of artificial intelligence implementation. The previous chapter discussed a sublanguage in news stories used to introduce statements made by market players. This sublanguage includes repetitive patterns of person and organisation names which appear before and after these statements. The challenge here is to discover a technique that is capable of dealing with these patterns which can be used to extract associated proper names.

This chapter presents the tool Nextract, currently being developed as an archetype for local grammars based on a Named Entity extraction system. We first discuss the system specification concentrating on the main challenges during the system development. This is followed by the system design and functionality of NExtract components. Then we explain the implementation in Java and justify the reasons for choosing this programming language. We also show, using a series of screenshots, the procedures required to run NExtract.

We also discuss the evaluation strategy adopted and the encouraging results, which have been obtained.

4.1 System Specification

Figure 4.1 shows an overview of NExtract, which demonstrates the extraction of proper names from news texts. The lexical analysis and sublanguage parsing are the two essential stages of the extraction process. These two challenges guided us during the development of NExtract. On one

hand, there were considerations about achieving flexible and good quality lexical analysis results, while on the other hand, there were issues relating to the implementation of the sublanguage parser which guarantee an easy and quick revision of the grammars when required.

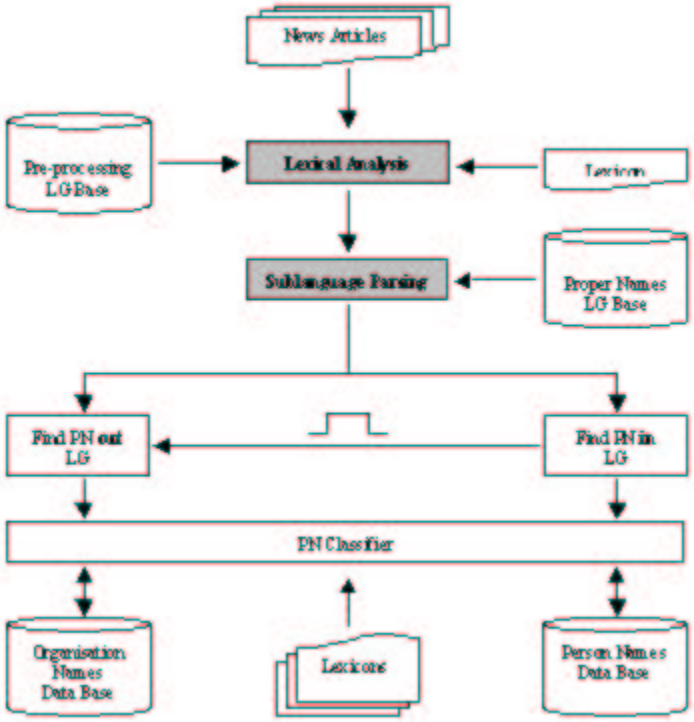

Figure 4.1: Overview of NExtract.

4.1.1 Lexical Analysis

The lexical analysis includes tasks like normalisation, tokenisation, Part-of-Speech (POS) tagging and the splitting of input texts into sentences. The task of splitting texts into sentences is vital for proper names analysis. Person names may be preceded or have an abbreviated form, e.g. *George W. Bush*. Organisation names may appear in abbreviations e.g. *N.A.T.O*. One way to accomplish this task is to submit the text to a finite-state pre-processing component that puts the mark {*S*} between each sentence [41]. The lexical analysis component incorporated in NExtract borrows programs from the Unitex[13] system. These programs are implemented in C and can easily be called from a Java interface [42].

[13] This software is based on a concept born at LADL (Laboratoire d'Automatique Documentaire et Linguistique), under the direction of its director, Maurice Gross.

4.1.2 Sublanguage Parsing

This unit draws on a cascade of Finite State Transducers (FST) obtained from the analysis conducted in section 3.2.1. These grammars are first compiled into FSTs and then applied to the input text using the graph editor and some C-based programs provided with Unitex.

4.2 System Architecture

A diagram of our named entity extraction system is shown in Figure 4.1.The detailed functionality of each module is discussed here.

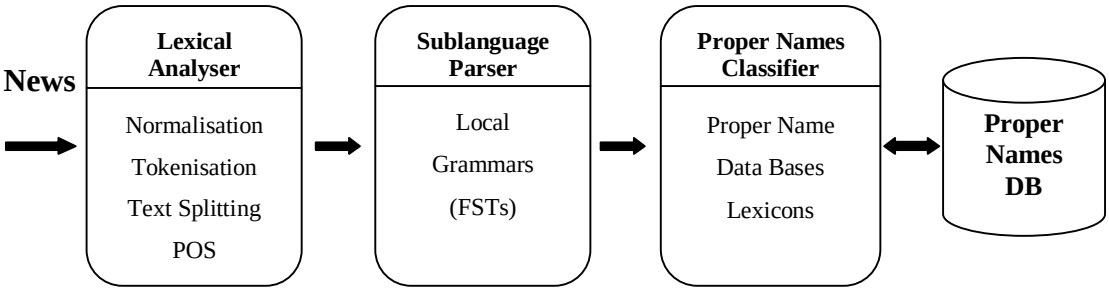

Figure 4.1: The architecture of NExtract.

4.2.1 Lexical Analyser

The lexical analyser is responsible for:

- Normalising consecutive separators. That is, a sequence of white spaces is replaced by only one space and all other sequence of separators containing at least one new line are normalised into a unique new line.
- Splitting a document into sentences.
- Breaking sentences into tokens.
- Looking up the tokens in dictionaries.
- Optionally assigning syntactic categories, such as part-of-speech tags, to the tokens.

4.2.2 Sublanguage Parser

The sublanguage parser forms the heart of our system. It uses the automata-oriented technology to find and isolate local grammars previously compiled into transducers.

Transducers are finite-state machines with an alphabet input and output. The input alphabet holds patterns to be recognised in texts whereas the output alphabet contains, in our case, information marked out in a language that serves the proper names classification stage. The patterns we are looking for are those discussed in Chapter 3. Below is an example of a pattern found in a text and marked out by the transducer cascade:

said Steve Russell, UK equity strategist at HSBC Securities. $\Rightarrow$

**** said <Steve Russell>_PN , <UK equity strategist>_TI at <HSBC Securities>_ORG . ***{S}*

The stars at the beginning and end of the pattern help define its boundaries. The *PN*, *TI* and *ORG* are used to reflect person, professional title and organisation names respectively.

4.2.3 Proper Names Classifier

The classifier is based on two fundamental observations:

1. The classification accuracy of names identified by local grammars is quite high.
2. The authors of news articles first give the canonical form of person and organisation names, and then abbreviated forms [43].

Therefore, NExtract first populates the databases of proper names with names found by local grammars. These databases together with lexicons of English first names and organisation markers are then used to disambiguate names appearing out of local grammars.

4.3 System Implementation

NExtract is mainly written in Java and implemented under Java 1.4.1_01. It borrows some C programs from Unitex system to take care of the lexical analysis and sublanguage parsing. This option assured a tested, reliable and fast text pre-processing and finite-state engine. Additionally, Java offers a powerful regular expression engine, which we use to tackle names appearing out of local grammars. The graphical user interface of NExtract is written in Java for its ease of use and quick prototyping. In addition, the Java implementation of NExtract has another number of significant technical advantages. These include portability, object-orientation, robustness, and scripting language friendliness.

4.4 Walking through NExtract

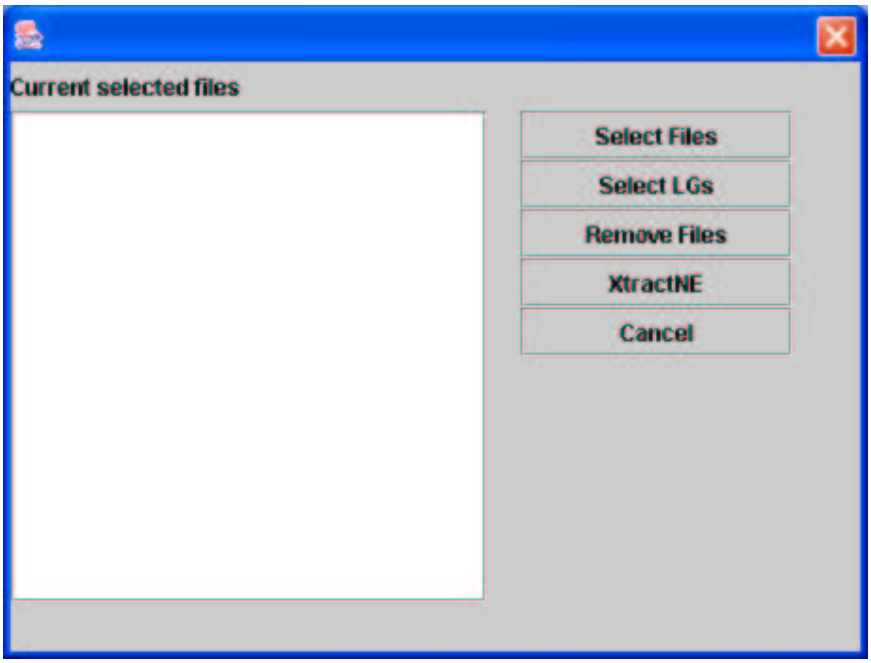

Figure 4.2: NExtract main window.

Running NExtract is a simple process with the assistance of the graphical user interface, which shows clearly the proceeding steps (see Figure 4.2). In the first step, users can select one or many news files and then click the button "Open" (see Figure 4.3).

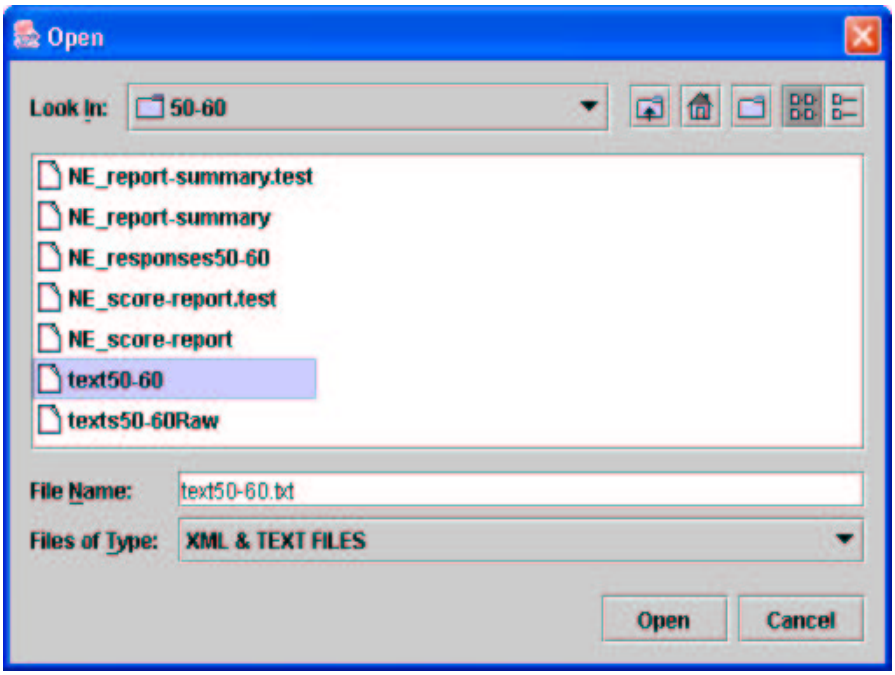

Figure 4.3: NExtract open file window.

Having decided on the file(s) to be processed, users may select which local grammars the system should use to locate proper names. This is achieved by pressing the button labelled "Select LGs". Once that is done, a 'choose file' window is opened (see Figure 4.4).

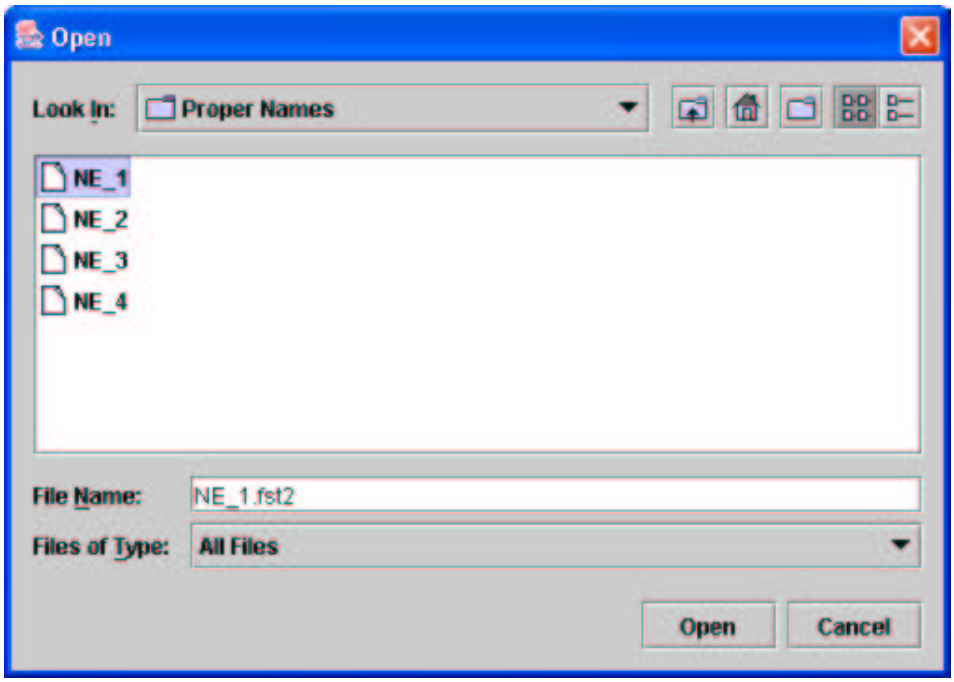

Figure 4.4 : The stage of choosing local grammars.

If needed, the button labelled "Remove Files" can be used to remove as many files as the user needs from the list of files already defined to be processed. The final step to go through is to press the button "XtractNE" and wait for the result windows (see Figure 4.5 and Figure 4.6) that show person and organisation names together with their numbers of appearance in the input file(s).

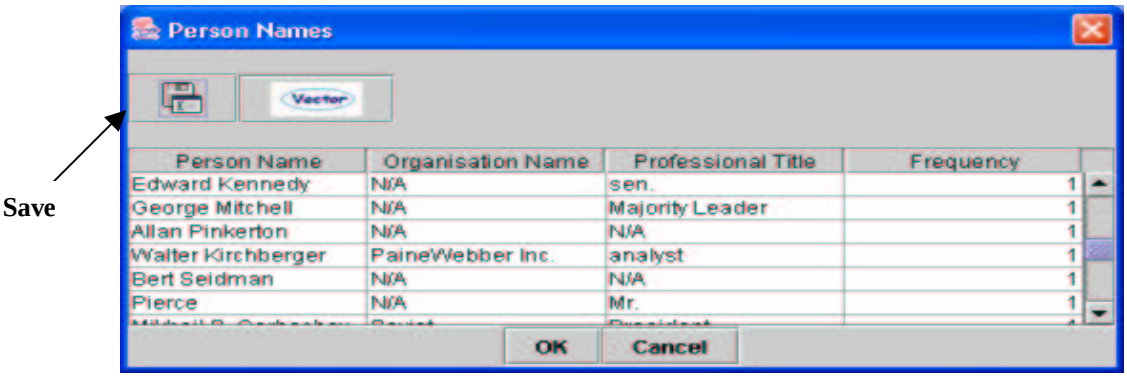

Figure 4.5: The window person name results.

The window of person names shows two additional fields (Organisation Name and Professional Title). These fields may contain information about the organisation the

identified person works for as well as his or her role in that organisation. Otherwise they are filled in with N/A (Not Applicable).

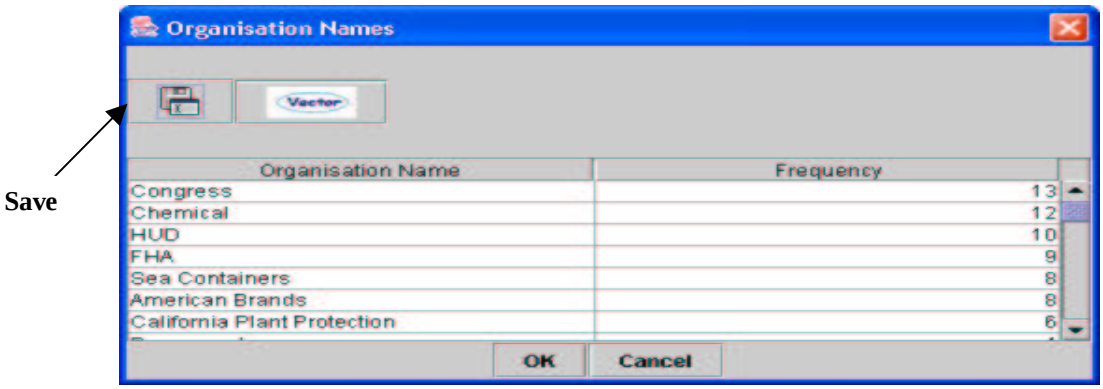

Figure 4.6: The window of organisation name results.

Finally, proper names extracted by NExtract can be saved in XML format using the buttons pointed to by the arrows in both person and organisation name result windows. The format devised to store person names is shown in Figure 4.7.

```
- <Tuple>
- <field name="Person Name">
<value>Walter Yetnikoff</value>
</field>
- <field name="Organisation Name">
<value>CBS Records</value>
</field>
- <field name="Professional Title">
<value>Chairman</value>
</field>
- <field name="Frequency">
<value>1</value>
</field>
</Tuple>
```

Figure 4.7: Example of a person name saved in XML format.

4.5 System Evaluation

Evaluation is the process of examining a system or system component to determine the extent to which specified properties are present. Thus, the evaluation of NExtract can help us: (a) verify how successfully the sublanguage approach is implemented in NExtract and (b) validate the hypothesis that there are idiosyncratic and recurring patterns in the language used when journalists introduce the speech of influential personalities.

The following subsections cover the evaluation strategy used to evaluate NExtract, the details of how the evaluation is carried out and the evaluation results.

4.5.1 Evaluation Strategy

Evaluation strategy is usually established after answering a series of questions that may seek the motivation of the evaluation and what we hope to discover through the evaluation ([44] as seen in [19]).

The motivation for evaluating NExtract is to investigate the scientific justification of the hypothesis we have made in respect to the sublanguage of proper names in news articles. Through the evaluation, we hope to determine how NExtract's output compares with the output of human linguists in tagging person and organisation names. The MUC scoring software is used for that comparison. This software reads in an output file produced by a linguist, containing *keys*, and an output file produced by a software system, containing *responses*. Having aligned objects in the key file with objects in the response file, the scorer then calculates various scores based on how well the responses agree with the keys [45]. Figure 4.8 sketches the scoring process adopted by the Message Understanding Conferences (MUCs).

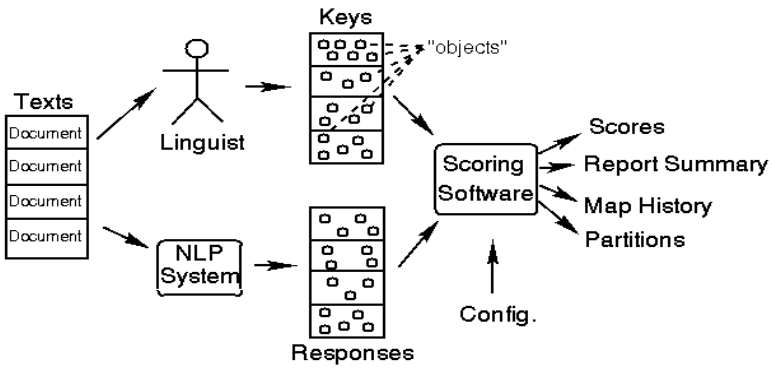

Figure 4.8: The system architecture of the MUC scoring software [45].

In our evaluation, we use news articles compiled from both the Wall Street Journal and Reuters. MUC primary measures are used: *Precision* and *Recall*.

Precision is the measure of how much of the response fills are actually in the key.

$$PRE = \frac{COR + (0.5 * PAR)}{ACT}$$

Recall is the measure of how much of the key fills were produced in the response.

$$REC = \frac{COR + (0.5 * PAR)}{POS}$$

where

COR: Correct

The two single fills are considered identical.

PAR: Partially Correct

The two single fills are not identical, but partial credit should still be given.

ACT: Actual

The number of fills in the response.

POS: Possible

The number of fills in the key, which contribute to the final score.

However, responses from a system tuned for high recall may differ from those obtained from a system tuned for high precision. This fact promotes the use of van Rijsbergen's F-measure, which is a combination of both recall and precision measures. The formula for F is:

$$F = \frac{(\beta^2 + 1) * P * R}{(\beta^2 * P) + R}$$

where β is the relative weight of precision and recall.

4.5.2 Evaluation in Action

Having decided on the method, the evaluation can be put into action. Two test beds are designed to contain 24 sets of news articles with ten texts each. The first test bed contains news articles randomly chosen from the Reuters corpus (2002) and manually annotated by a group of PhD students in the Computing Department of the University of Surrey. The texts in the second test bed are arbitrarily taken from the corpus of the sixth MUC, amassed from Wall Street Journal (WSJ) and manually annotated by the Language Data Consortium (LDC).

The specific writing style for one source of news articles, for instance Reuters, often makes certain the existence of consistent patterns within the same source. The Reuters test sets allow us to evaluate the performance of NExtract against the same data upon which our text analysis was based, whereas the WSJ test sets provide the test material on which NExtract was not based, thereby testing the extensibility of NExtract . At the same

time, the robustness of our system and ability to handle unseen inputs can also be evaluated.

4.5.3 Evaluation Results

NExtract was tested against two different news text sources - Reuters and Wall Street Journal. The performance of our system has been measured by the scale of precision and recall. The results against Reuters' data sets are presented in Table 4.1.

Table 4.1: *Precision* and *Recall* of NExtract against Reuters data sets.

Set	**Reuters**		
	$R_{Reuters}$	$P_{Reuters}$	$F_{Reuters}$
1	88%	92%	90%
2	87%	89%	88%
3	90%	90%	90%
4	90%	90%	89.61%
5	93%	87%	89.7%
6	81%	89%	85.6%
7	79%	94%	85.9%
8	89%	93%	90.8%
9	92%	91%	91.4%
10	93%	95%	94%
11	92%	95%	93.6%
12	87%	93%	90%
Average	**87.92%**	**90.92%**	**89.31%**

Out of the twelve sets of Reuters' data, the best performance was recorded on Sets 10 and 11 (94% and 93.6% respectively). This is hardly surprising since the analysis of local grammars was based on a set of news articles where these two sets were originally compiled, that of the training corpus introduced in Chapter 3.

The lowest F-measure (85.6%) for Set 6 of Reuters data when compared to other sets of the same source is largely due to the fact that most of the included person and organisation names appear out of the set of local grammars we have compiled so far, or are not accompanied with any clues (internal and external evidences) that would help their correct recognition. Names of one word length and satisfy the latter case are marked out as proper names if they do not belong to the DELAF simple word dictionary provided with Unitex. Consider the organisation name *Yell* (Britain's biggest phone directory publisher). This word has two definitions in DELAF dictionary

(yell,.V:W:P1s:P2s:P1p:P2p:P3p and yell,.N:s) and appears in the local grammars 15 times in one of the files of Set 6. Hence, the dramatic decrease in the recall measure (49%) of that file which, in turn, deteriorates the F-measure results of the whole of Set 6.

Table 4.2: *Precision* and *Recall* of NExtract against WSJ data sets.

Set	**Wall Street Journal**		
	R_{WSJ}	P_{WSJ}	F_{WSJ}
1	92%	80%	85.6%
2	86%	85%	85.5%
3	80%	80%	80.0%
4	87%	77%	81.7%
5	80%	79%	79.5%
6	89%	77%	82.6%
7	77%	77%	77.0%
8	87%	80%	83.4%
9	74%	78%	75.9%
10	86%	69%	76.6%
11	61%	65%	62.9%
12	74%	65%	69.2%
Average	**81.1%**	**76%**	**78.3%**

Table 4.2 shows the results of NExtract when tested on sets of texts published in 1989 by the Wall Street Journal. Despite the small drop (11%) of performance against these sets in comparison to Reuters, our local grammars are found to live in entirely novel inputs. This fact provides strong supporting evidence for our hypothesis on the existence of the *distinctive and repetitive patterns of proper names*.

Detailed results obtained for individual news stories included in Set 1 of Reuters and Set 2 of the Wall Street Journal are shown in Appendices A and B respectively.

Chapter 5

5 Conclusion and Future Work

In this thesis we have discussed and explored many aspects of proper names including their onomastic behaviour, linguistic characteristics and more significantly, the computational work that has been carried out in the analysis of proper names to date. In Chapter 3 we analysed English news articles and studied the language used by journalists to introduce person and organisation names. Our study confirmed the existence of a sublanguage within the contexts of these classes of names. This theoretical claim is examined by means of the construction of a computational archetype, NExtract, to extract proper names from natural language texts such as news articles (Chapter 4).

This archetype includes three main functional units. The pre-processing unit prepares the input data and provides it with syntactic information. The sublanguage parser marks out names occurring in local grammars. Finally, names are classified by a proper name classifier which first searches for names already identified by the sublanguage parser and stores them in an intermediate template. It then exploits this template together with lists of internal and external evidences to disambiguate names emerging out of local grammars. This modular architecture was designed with the capacity of extension to other domains that operate local grammars . However, this research is presently focused on the automatic extraction of proper names from news article texts.

In this chapter we conclude the thesis by suggesting that local grammars can be used as an effective approach to increase the accuracy of a proper name analysis system. Finally, we propose further directions to be explored in order take our research further.

5.1 Contributions

This thesis considers the analysis of proper names in unedited, naturally occurring texts. One of the main obstacles to such analysis is the poor coverage of proper names in

conventional dictionaries. Concerning this point we have demonstrated that poor coverage in dictionaries or in system lexica need not be a problem as proper names very often appear in local grammars in the text in which they occur. This means that we can identify the unknown proper names through an analysis of the accompanying context.
One major source of information relating to proper names is biographical dictionaries. The entries in this kind of dictionary only cover names of such famous people as monarchs, national leaders, and scientists. Our contribution has been to show that news articles can also be a reliable source from which we can acquire information about less prominent figures such as organisation executives and officers.

The preliminary work on proper name extraction is regarded as very promising for automatic compilation from large text corpora. Our work could be of great assistance in the production of printed lexica of proper names and encyclopaedias.

Additional contributions within the field of proper name extraction have been made during the course of this research, including the design and implementation of a local grammar-based proper name extraction system, as described in Chapter 4.

An evaluation was performed to compare the system output for person and organisation names with those identified by expert linguists of LDC (Section 4.5). The extraction accuracy of these classes of names achieved by the system was about 90% for Reuters' texts and 78% for LDC texts as compared to humans. This has laid the foundation for using local grammars to recognise proper names in news article texts and further improvements are possible.

5.2 Future Work

The encouraging results obtained with the adoption of the local grammar notion to identify proper names in natural language texts will motivate further investigation. Our future work plan remains busy. Its agenda is mainly to explore further the theoretical roots of local grammar and applicable aspects of proper names.

On a theoretical level, we intend to examine Harrisian transformational grammar as well as the role of finite state automata/transducers in proper name analysis. A good grasp of the latter will assist the implementation process, while the former will help us learn more about the representation of proper name local grammars in different languages and domains. We are also keen to extend our local grammar approach to French and Arabic to examine whether it operates across languages. If it does, then one may conclude that proper names are governed by their own grammar.

At the application level, we have three main targets. The first concerns the analysis of news articles, especially sentiment analysis where attribution of the sentiment person or organisation makes a difference in financial markets. The second focuses on integrating our local grammar of proper names in a computational system that creates candidate entries for a simple biographical dictionary. The third is to expand the coverage of our analysis to deal with proper names other than those we focus on for the time being: person and organisation names. Place names are amongst those on our priority list.

As of 1 October 2004, I have twelve more months to go for my PhD. Table 5.1 proposes the timing of future tasks required to complete my PhD by Oct 2005.

Table 5.1: Timing of future tasks for the completion of the PhD.

Task #	Task Name	2004			2005								
		Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep
1	To investigate the theory of grammar transformation												
2	To explore ways to attribute market sentiments												
3	To extend system coverage												
4	To extend work to French and Arabic												
5	Thesis Write-up												

Appendix A

Evaluation Results for NExtract on Reuters Texts

	P&R	2P&R	P&2R
F-MEASURES	94.07	94.66	93.49

01Oct2002_0717 key: 01Oct2002_0717 response: 01Oct2002_0717

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
enamex	7	9	7	0	0	0	2	0	100	78	0	22	0	22
type	7	9	7	0	0	0	2	0	100	78	0	22	0	22
text	7	9	7	0	0	0	2	0	100	78	0	22	0	22
timex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
numex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL	14	18	14	0	0	0	4	0	100	78	0	22	0	22

07Oct2002_1611 key: 07Oct2002_1611 response: 07Oct2002_1611

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
enamex	11	11	11	0	0	0	0	0	100	100	0	0	0	0
type	11	11	10	0	1	0	0	0	91	91	0	0	9	9
text	11	11	10	0	1	0	0	0	91	91	0	0	9	9
timex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
numex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL	22	22	20	0	2	0	0	0	91	91	0	0	9	9

08Oct2002_0913 key: 08Oct2002_0913 response: 08Oct2002_0913

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
enamex	40	38	38	0	0	2	0	0	95	100	5	0	0	5
type	40	38	38	0	0	2	0	0	95	100	5	0	0	5
text	40	38	37	0	1	2	0	0	93	97	5	0	3	8
timex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
numex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL	80	76	75	0	1	4	0	0	94	99	5	0	1	6

15Oct2002_0132 key: 15Oct2002_0132 response: 15Oct2002_0132

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
enamex	39	37	37	0	0	2	0	0	95	100	5	0	0	5
type	39	37	37	0	0	2	0	0	95	100	5	0	0	5
text	39	37	36	0	1	2	0	0	92	97	5	0	3	8
timex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
numex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL	78	74	73	0	1	4	0	0	94	99	5	0	1	6

17Oct2002_0723 key: 17Oct2002_0723 response: 17Oct2002_0723

```
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+-------------+-----------------------
enamex           3     2|   2   0    0|   1    0    0|  67 100  33   0   0  33
  type           3     2|   2   0    0|   1    0    0|  67 100  33   0   0  33
  text           3     2|   2   0    0|   1    0    0|  67 100  33   0   0  33
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+-------------+-----------------------
TOTAL            6     4|   4   0    0|   2    0    0|  67 100  33   0   0  33
------------------------+-------------+-------------+-----------------------

21Oct2002_0858 key: 21Oct2002_0858 response: 21Oct2002_0858
------------------------+-------------+-------------+-----------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+-------------+-----------------------
enamex          22    19|  19   0    0|   3    0    0|  86 100  14   0   0  14
  type          22    19|  19   0    0|   3    0    0|  86 100  14   0   0  14
  text          22    19|  19   0    0|   3    0    0|  86 100  14   0   0  14
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+-------------+-----------------------
TOTAL           44    38|  38   0    0|   6    0    0|  86 100  14   0   0  14
------------------------+-------------+-------------+-----------------------

23Oct2002_1706 key: 23Oct2002_1706 response: 23Oct2002_1706
------------------------+-------------+-------------+-----------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+-------------+-----------------------
enamex          26    26|  25   0    0|   1    1    0|  96  96   4   4   0   7
  type          26    26|  25   0    0|   1    1    0|  96  96   4   4   0   7
  text          26    26|  25   0    0|   1    1    0|  96  96   4   4   0   7
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+-------------+-----------------------
TOTAL           52    52|  50   0    0|   2    2    0|  96  96   4   4   0   7
------------------------+-------------+-------------+-----------------------

29Oct2002_1745 key: 29Oct2002_1745 response: 29Oct2002_1745
------------------------+-------------+-------------+-----------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+-------------+-----------------------
enamex          17    17|  17   0    0|   0    0    0| 100 100   0   0   0   0
  type          17    17|  16   0    1|   0    0    0|  94  94   0   0   6   6
  text          17    17|  17   0    0|   0    0    0| 100 100   0   0   0   0
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+-------------+-----------------------
TOTAL           34    34|  33   0    1|   0    0    0|  97  97   0   0   3   3
------------------------+-------------+-------------+-----------------------

30Oct2002_0956 key: 30Oct2002_0956 response: 30Oct2002_0956
------------------------+-------------+-------------+-----------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+-------------+-----------------------
enamex          21    21|  21   0    0|   0    0    0| 100 100   0   0   0   0
  type          21    21|  21   0    0|   0    0    0| 100 100   0   0   0   0
  text          21    21|  21   0    0|   0    0    0| 100 100   0   0   0   0
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+-------------+-----------------------
TOTAL           42    42|  42   0    0|   0    0    0| 100 100   0   0   0   0
------------------------+-------------+-------------+-----------------------
```

```
31Oct2002_2216 key: 31Oct2002_2216 response: 31Oct2002_2216
------------------------+-------------+-------------+-----------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
```

enamex	10	12	10	0	0	0	2	0	100	83	0	17	0	17
type	10	12	8	0	2	0	2	0	80	67	0	17	20	33
text	10	12	8	0	2	0	2	0	80	67	0	17	20	33
timex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
numex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TOTAL	20	24	16	0	4	0	4	0	80	67	0	17	20	33

* * * TOTAL SLOT SCORES * * *

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
enamex	196	192	187	0	0	9	5	0	95	97	5	3	0	7
type	196	192	183	0	4	9	5	0	93	95	5	3	2	9
text	196	192	182	0	5	9	5	0	93	95	5	3	3	9
timex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
type	0	0	0	0	0	0	0	0	0	0	0	0	0	0
text	0	0	0	0	0	0	0	0	0	0	0	0	0	0
numex	0	0	0	0	0	0	0	0	0	0	0	0	0	0
type	0	0	0	0	0	0	0	0	0	0	0	0	0	0
text	0	0	0	0	0	0	0	0	0	0	0	0	0	0
ALL OBJECTS	392	384	365	0	9	18	10	0	93	95	5	3	2	9

* * * TASK SUBCATEGORIZATION SCORES * * *

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
Enamex:														
organizati	161	160	152	0	3	6	5	0	94	95	4	3	2	8
person	35	32	31	0	1	3	0	0	89	97	9	0	3	11
location	0	0	0	0	0	0	0	0	0	0	0	0	0	0
other	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Timex:														
date	0	0	0	0	0	0	0	0	0	0	0	0	0	0
time	0	0	0	0	0	0	0	0	0	0	0	0	0	0
other	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Numex:														
money	0	0	0	0	0	0	0	0	0	0	0	0	0	0
percent	0	0	0	0	0	0	0	0	0	0	0	0	0	0
other	0	0	0	0	0	0	0	0	0	0	0	0	0	0

* * * DOCUMENT SECTION SCORES * * *

SLOT	POS	ACT	COR	PAR	INC	MIS	SPU	NON	REC	PRE	UND	OVG	SUB	ERR
HEADLINE	0	0	0	0	0	0	0	0	0	0	0	0	0	0
headline	0	0	0	0	0	0	0	0	0	0	0	0	0	0
HL	0	0	0	0	0	0	0	0	0	0	0	0	0	0
hl	0	0	0	0	0	0	0	0	0	0	0	0	0	0
SLUG	0	0	0	0	0	0	0	0	0	0	0	0	0	0
PREAMBLE	0	0	0	0	0	0	0	0	0	0	0	0	0	0
DD	0	0	0	0	0	0	0	0	0	0	0	0	0	0
DATE	0	0	0	0	0	0	0	0	0	0	0	0	0	0
DATELINE	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TXT	392	384	365	0	9	18	10	0	93	95	5	3	2	9
TEXT	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IDS	0	0	0	0	0	0	0	0	0	0	0	0	0	0
TRAILER	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Appendix B

Evaluation Results for NExtract on MUC Texts

```
                                                        P&R      2P&R      P&2R
F-MEASURES                                            85.67     85.51     85.83

8911010051     key: 8911010051     response:     8911010051
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex           7     5|   5   0    0|   2    0    0|  71 100  29   0   0  29
  type           7     5|   5   0    0|   2    0    0|  71 100  29   0   0  29
  text           7     5|   5   0    0|   2    0    0|  71 100  29   0   0  29
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           14    10|  10   0    0|   4    0    0|  71 100  29   0   0  29
------------------------+-------------+--------------+------------------------

8911010062     key: 8911010062     response:     8911010062
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex           3     3|   3   0    0|   0    0    0| 100 100   0   0   0   0
  type           3     3|   3   0    0|   0    0    0| 100 100   0   0   0   0
  text           3     3|   3   0    0|   0    0    0| 100 100   0   0   0   0
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL            6     6|   6   0    0|   0    0    0| 100 100   0   0   0   0
------------------------+-------------+--------------+------------------------

8911010080     key: 8911010080     response:     8911010080
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex          54    55|  51   0    0|   3    4    0|  94  93   6   7   0  12
  type          54    55|  48   0    3|   3    4    0|  89  87   6   7   6  17
  text          54    55|  46   0    5|   3    4    0|  85  84   6   7  10  21
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL          108   110|  94   0    8|   6    8    0|  87  85   6   7   8  19
------------------------+-------------+--------------+------------------------

8911010084     key: 8911010084     response:     8911010084
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex          13    15|  12   0    0|   1    3    0|  92  80   8  20   0  25
  type          13    15|  12   0    0|   1    3    0|  92  80   8  20   0  25
  text          13    15|  12   0    0|   1    3    0|  92  80   8  20   0  25
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           26    30|  24   0    0|   2    6    0|  92  80   8  20   0  25
------------------------+-------------+--------------+------------------------
```

```
8911010104     key: 8911010104     response:     8911010104
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex          15    14|  13   0    0|   2    1    0|  87  93  13   7   0  19
  type          15    14|  13   0    0|   2    1    0|  87  93  13   7   0  19
  text          15    14|  13   0    0|   2    1    0|  87  93  13   7   0  19
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           30    28|  26   0    0|   4    2    0|  87  93  13   7   0  19
------------------------+-------------+--------------+------------------------
8911010119     key: 8911010119     response:     8911010119
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex           5     6|   5   0    0|   0    1    0| 100  83   0  17   0  17
  type           5     6|   5   0    0|   0    1    0| 100  83   0  17   0  17
  text           5     6|   5   0    0|   0    1    0| 100  83   0  17   0  17
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           10    12|  10   0    0|   0    2    0| 100  83   0  17   0  17
------------------------+-------------+--------------+------------------------

8911010138     key: 8911010138     response:     8911010138
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex           9     8|   8   0    0|   1    0    0|  89 100  11   0   0  11
  type           9     8|   8   0    0|   1    0    0|  89 100  11   0   0  11
  text           9     8|   8   0    0|   1    0    0|  89 100  11   0   0  11
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           18    16|  16   0    0|   2    0    0|  89 100  11   0   0  11
------------------------+-------------+--------------+------------------------

8911010156     key: 8911010156     response:     8911010156
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex          19    18|  17   0    0|   2    1    0|  89  94  11   6   0  15
  type          19    18|  17   0    0|   2    1    0|  89  94  11   6   0  15
  text          19    18|  16   0    1|   2    1    0|  84  89  11   6   6  20
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           38    36|  33   0    1|   4    2    0|  87  92  11   6   3  18
------------------------+-------------+--------------+------------------------

8911010174     key: 8911010174     response:     8911010174
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex          22    23|  18   0    0|   4    5    0|  82  78  18  22   0  33
  type          22    23|  17   0    1|   4    5    0|  77  74  18  22   6  37
  text          22    23|  16   0    2|   4    5    0|  73  70  18  22  11  41
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           44    46|  33   0    3|   8   10    0|  75  72  18  22   8  39
------------------------+-------------+--------------+------------------------

8911020018     key: 8911020018     response:     8911020018
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex          13    14|  13   0    0|   0    1    0| 100  93   0   7   0   7
  type          13    14|  13   0    0|   0    1    0| 100  93   0   7   0   7
  text          13    14|  10   0    3|   0    1    0|  77  71   0   7  23  29
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TOTAL           26    28|  23   0    3|   0    2    0|  88  82   0   7  12  18
------------------------+-------------+--------------+------------------------
```

```
 * * * TOTAL SLOT SCORES * * *
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
enamex         160   161| 145   0    0|  15   16    0|  91  90   9  10   0  18
  type         160   161| 141   0    4|  15   16    0|  88  88   9  10   3  20
  text         160   161| 134   0   11|  15   16    0|  84  83   9  10   8  24
timex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  type           0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  text           0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
numex            0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  type           0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  text           0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
ALL OBJECTS    320   322| 275   0   15|  30   32    0|  86  85   9  10   5  22
------------------------+-------------+--------------+------------------------

 * * * TASK SUBCATEGORIZATION SCORES * * *
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
Enamex:
  organizati   104   109|  92   0    2|  10   15    0|  88  84  10  14   2  23
  person        56    52|  49   0    2|   5    1    0|  88  94   9   2   4  14
  location       0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  other          0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
Timex:
  date           0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  time           0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  other          0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
Numex:
  money          0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  percent        0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
  other          0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------

 * * * DOCUMENT SECTION SCORES * * *
------------------------+-------------+--------------+------------------------
SLOT           POS   ACT| COR PAR INC | MIS  SPU  NON| REC PRE UND OVG SUB ERR
------------------------+-------------+--------------+------------------------
HEADLINE         0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
headline         0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
HL              30    18|  16   0    2|  12    0    0|  53  89  40   0  11  47
hl               0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
SLUG             0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
PREAMBLE         0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
DD               0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
DATE             0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
DATELINE         2     0|   0   0    0|   2    0    0|   0   0 100   0   0 100
------------------------+-------------+--------------+------------------------
TXT            288   304| 259   0   13|  16   32    0|  90  85   6  11   5  19
TEXT             0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
IDS              0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
------------------------+-------------+--------------+------------------------
TRAILER          0     0|   0   0    0|   0    0    0|   0   0   0   0   0   0
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